

NEO InventoriesTM

for the

NEO Personality Inventory-3TM (NEO-PI-3TM)

NEO Five-Factor Inventory-3TM (NEO-FFI-3TM)

NEO Personality Inventory-RevisedTM (NEO PI-RTM)

Professional Manual

Robert R. McCrae, PhD

Paul T. Costa, Jr., PhD

PAR[®]

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NEO Personality Inventory-Revised™ (NEO PI-R™)

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Acknowledgments

In the quarter century since the introduction of the NEO Inventories, psychologists have accumulated a wealth of data on the instruments' validity and experience with their utility in clinical, applied, and research contexts. The value of the inventories has grown because of these contributions, and we are grateful for them, as we are for the critiques that have stimulated our own research and revisions. It is no longer possible to list all those who have made significant contributions, but we are particularly thankful for the work of Jüri Allik, R. Michael Bagby, Spencer R. Baker, Filip De Fruyt, William W. Eaton, Kerry Jang, John E. Kurtz, Gerald Nestadt, Ralph L. Piedmont, Jean Pierre Rolland, Ilene C. Siegler, Antonio Terracciano, Thomas A. Widiger, Jerry Wiggins, Redford Williams, and Jian Yang. Thomas A. Martin and his students showed the utmost professionalism in the collection of data for the item selection and validation of the NEO Personality Inventory-3. We deeply appreciate the efforts of collaborators around the world who have translated the NEO Inventories into over 50 languages and have provided data documenting the universality of traits in the Five-Factor Model. We also thank the research participants whose honest and careful responses are an essential foundation of trait psychology. Finally, we acknowledge Bill and Karol for their steadfast support.

Robert R. McCrae
Paul T. Costa, Jr.
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From the 1992 Edition

In a very real sense, the Revised NEO Personality Inventory (NEO PI-R) is the result of the work of many personality psychologists and psychometricians who preceded us, especially those whose work led to the Five-Factor Model of personality. We are indebted to them, as we are to the many volunteers who participated in our research. In addition, we would like to express our appreciation to the following psychologists who provided normative data, case studies, translations, and other assistance: Alois Angleitner, University of Bielefeld; Michael Harris Bond, Chinese University of Hong Kong; Peter Borkenau, University of Bielefeld; Philippe Cappeliez, University of Ottawa; Lee Anna Clark, Southern Methodist University; Theodore Dembroski, University of Maryland, Baltimore County; John M. Digman, Oregon Research Institute; Peter Fagan, Johns Hopkins University; Lewis R. Goldberg, University of Oregon; Hans Hoekstra, PTT/Research, Groningen, The Netherlands; Wai-kwan Li, Chinese University of Hong Kong; Brian Little, Carleton University; James MacDougall, Eckerd College; Timothy Miller, Stockton, California; Inger Ulrikke Møller, University of Oslo; I. Montag, Tel Aviv University; Erik Mutén, Montague Psychological Associates; Fritz Ostendorf, University of Koblenz-Landau; Ilene Siegler, Duke University Medical School; and David Watson, Southern Methodist University.

From the 1989 Edition

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Introduction

The NEO Inventories are concise measures of the five major dimensions, or domains, of personality and the most important traits or facets that define each domain. Together, the five broad domain scales and the 30 specific facet scales allow a comprehensive assessment of adolescent and adult personality.

The NEO Inventories comprise a series of closely related instruments developed over a period of 30 years from the original NEO Inventory, a measure of 18 traits grouped into the domains of Neuroticism (N), Extraversion (E), and Openness to Experience (O). Since 1985, the Inventories have also included measures of two other basic dimensions of personality, Agreeableness (A) and Conscientiousness (C). This Professional Manual describes the current versions of the NEO Inventories: The NEO Personality Inventory-3 (NEO-PI-3); the Revised NEO Personality Inventory (NEO PI-R); and the short form, the NEO Five-Factor Inventory-3 (NEO-FFI-3).

The major innovation in this edition is the publication of the NEO-PI-3, an instrument suitable for the assessment of individuals age 12 years and older. It is a modification of the NEO PI-R in which 37 of the 240 items have been replaced. The new items are easier to understand and have better psychometric properties. In addition, the NEO-PI-3 has been shown to be suitable for assessing personality in middle school-aged children and adolescents, as well as in adults. The NEO-PI-3 retains the validity and interpretation of the NEO PI-R, but is applicable to a wider range of respondents, including adults with lower educational levels. This Professional Manual includes new normative data for the NEO-PI-3.

Researchers and clinicians who have extensive experience with the NEO PI-R may wish to continue using that version and the norms with which they are familiar. For that reason, the NEO PI-R will continue to be made available in both print and computer versions.

This Professional Manual also introduces the NEO-FFI-3. This 60-item instrument assesses the five major domains of personality. The NEO-FFI-3 is a revision of the NEO-FFI in which 15 of the 60 items have been replaced to improve

readability and psychometric properties. All items are taken from the NEO-PI-3; thus, the NEO-FFI-3 is also appropriate for respondents age 12 years and older. New norms are provided in this Professional Manual.

The NEO-FFI-3 supercedes the NEO-FFI. Researchers who need to use the original NEO-FFI (for example, for a longitudinal retest) must obtain it by license from PAR.

Among the other new features discussed in this Professional Manual are:

- the NEO Problems in Living Checklist (NEO-PLC), a tool for clinicians to assess problematic behaviors and symptoms associated with a NEO-PI-3 or NEO PI-R profile;
- the NEO Style Graph Booklet, a new way to provide feedback to respondents;
- the NEO Job Profiler, a tool to identify traits relevant to an occupation or position;
- adolescent norms for the observer rating version of the NEO-FFI-3;
- supplementary norms for different age groups and for international comparisons;
- a new item order that permits an abbreviated administration of the NEO-PI-3;
- a glossary of less familiar NEO-PI-3 words for respondents with limited literacy; and
- a list of published translations of the NEO Inventories.

There are two versions of all NEO Inventories: Form S, for self-reports, and Form R, for observer ratings. Form S consists of a series of items answered by respondents to describe themselves; Form R is a companion instrument with parallel items written in the third person for peer, spouse, or expert ratings. Form R can be used to obtain independent estimates on the same five domains of personality and may be of particular value in instances when it is desirable to validate or supplement self-reports.

The NEO Inventories embody a conceptual model that distills decades of factor analytic research on the structure

of personality. The scales themselves were developed and refined by a combination of rational and factor analytic methods and have been the subject of intensive research conducted for 30 years on both clinical and nonclinical samples. Evidence of scale reliability, stability, and construct validity has been presented in detail in a series of publications and is summarized in this Professional Manual. A bibliography of more than 2,500 publications using the NEO Inventories is available from the publisher's Web site, www.parinc.com.

Table 1 lists the NEO domains and their facets and provides a rough sense of the constructs measured by the instruments. To understand the constructs, professionals should

read chapter 5 on conceptualization and interpretation, review the correlates, which are discussed in chapter 8, and study the item content given in Appendix A of this Professional Manual.

The NEO Inventories are measures of general personality traits that have demonstrated their utility in clinical, applied, and research settings. This Professional Manual provides information on administration and scoring, the interpretation of scale scores and individual profiles, and suggested applications. It also summarizes the development and validation of the instruments and points to areas where further research is needed to widen understanding and application of the NEO Inventories.

Table 1
Domains and Facets Measured by the NEO Personality Inventory-3 (NEO-PI-3)

Domains	Extraversion (E) facets	Agreeableness (A) facets
N: Neuroticism	E1: Warmth	A1: Trust
E: Extraversion	E2: Gregariousness	A2: Straightforwardness
O: Openness	E3: Assertiveness	A3: Altruism
A: Agreeableness	E4: Activity	A4: Compliance
C: Conscientiousness	E5: Excitement-Seeking	A5: Modesty
Neuroticism (N) facets	E6: Positive Emotions	A6: Tender-Mindedness
N1: Anxiety	Openness (O) facets	Conscientiousness (C) facets
N2: Angry Hostility	O1: Fantasy	C1: Competence
N3: Depression	O2: Aesthetics	C2: Order
N4: Self-Consciousness	O3: Feelings	C3: Dutifulness
N5: Impulsiveness	O4: Actions	C4: Achievement Striving
N6: Vulnerability	O5: Ideas	C5: Self-Discipline
	O6: Values	C6: Deliberation

Test Materials and Use

NEO-PI-3 (and NEO PI-R) Materials

The NEO-PI-3 (and NEO PI-R) materials consist of this Professional Manual; reusable Form S and Form R Item Booklets; Hand-Scorable and/or Scannable Answer Sheets; six NEO-PI-3 Profile Forms; Your NEO Summary, which provides a description of the examinee's personality traits on the five factors; the NEO Style Graph Booklet, which assists in the interpretation of pairs of factor scores; and the NEO Problems in Living Checklist. In addition to the print materials, computerized components of the NEO-PI-3 are available. The measure can be computer-administered, scored, and interpreted or simply computer-scored and interpreted. The NEO Professional Report Service, a mail-in scoring and interpretation service, and the NEO On-Site Scanning Software require the use of the Scannable Answer Sheet, which is available from the publisher.

The NEO Software System—the computerized assessment system for the NEO Inventories—has its own On-Screen Help system. The computerized administration also gives instructions to the respondent. Most individuals, including those with no previous experience with computers, find this form of administration both easy and interesting.

Form S and Form R Item Booklets

There are two versions of the NEO-PI-3 Item Booklets: the self-report item booklet (i.e., Form S) and the observer-report item booklet (i.e., Form R). Both versions are appropriate for both adolescents and adults.

Form S is an eight-page booklet that contains the 240 statements that comprise the NEO-PI-3. The first page of the booklet presents the instructions for completing the Hand-Scorable Answer Sheet, and the second page presents the instructions for completing the Scannable Answer Sheet. The items of the NEO-PI-3 are presented on pages 3 through 8.

The Form R Item Booklet is parallel to the Form S Item Booklet, but with items written in the third person for peer,

spouse, or expert ratings. Form R-Female is to be used when the individual being rated is a girl or woman, and Form R-Male is to be used when the individual being rated is a boy or man. Like the Form S booklet, Form R booklets are eight pages long. The first page of each Form R booklet presents the instructions for completing the Hand-Scorable Answer Sheet, and the second page presents the instructions for completing the Scannable Answer Sheet. The items of the NEO-PI-3 are presented on pages 3 through 8.

Hand-Scorable (HS) and Scannable (SS) Answer Sheets

The Hand-Scorable (HS) Answer Sheet is a two-part, carbonless paper form designed for use with either Form S or Form R Item Booklets. (Although 37 of the items on the NEO-PI-3 differ from those of the NEO PI-R and the item order is different, the keying is identical, and thus the same Answer Sheet can be used for either instrument.) The HS Answer Sheet contains areas for recording basic demographic information and responses to the NEO-PI-3 items. The three validity check items of the NEO-PI-3 are also presented on the HS Answer Sheet. Respondents complete the demographic information and mark their item responses on the top sheet. This information is reproduced on the bottom sheet, which provides keys for scoring the domain and facet scales of the NEO-PI-3 (or the NEO PI-R).

The Scannable (SS) Answer Sheet is an 8½" x 11" form designed for use with either Form S or Form R Item Booklets. The front of the form presents instructions for completing the form. Additionally, the SS Answer Sheet contains an area for demographic information and directions for entering this information. The back of the form contains the area for recording responses to the NEO-PI-3 items. The three validity check items of the NEO-PI-3 are also presented on this side of the SS Answer Sheet. Completed SS Answer Sheets can be sent to the publisher for scoring and interpretation through the NEO Professional Report Service, or they can be used with the NEO-PI-3 On-Site Scanning Software.

Profile Forms

The NEO-PI-3 Profile Forms have been constructed to aid in interpretation. The Profile Forms allow raw scores to be easily converted to *T* scores for each domain and facet. The *T* scores presented on the forms have a mean of 50 and a standard deviation of 10.

There are six NEO-PI-3 Profile Forms; each form is color-coded and presents the five domains and 30 facets on an 11" x 17" sheet folded to 8½" x 11". Four Profile Forms offer male norms on one side and female norms on the other; two for Form S (i.e., one form for adolescents, one form for adults) and two for Form R (i.e., one form for adolescents, one form for adults). Two additional Profile Forms offer combined gender norms for adolescents (i.e., Form S on one side, Form R on the other side) and for adults (i.e., Form S on one side, Form R on the other side).

These Profile Forms are based on new normative data for the NEO-PI-3, which are provided in Appendix B of this Professional Manual. The five domain scale scores are plotted on the left-hand side of the form to give an overview of the individual's personality. The 30 facet scales, grouped by domain, follow. A graph of the profile may be drawn to visually portray the individual's scores.

Profile Forms for the NEO PI-R, based on the 1992 norms, are still available from the publisher. *Note that it is inappropriate to use the NEO PI-R Profile Forms to plot data from the NEO-PI-3; the new forms, which are based on new norms, are needed.*

NEO-FFI-3 Materials

The NEO-FFI-3 materials consist of this Professional Manual, the NEO-FFI-3 Form S and Form R Item Booklets, Your NEO Summary, and the NEO Style Graph Booklet. There are separate Item Booklets for adults and adolescents, and separate Form R Item Booklets for rating males and females. The Item Booklets are four-page, two-part carbonless forms. The first page presents instructions for completing the NEO-FFI-3, and the second page contains spaces for recording basic demographic information. The 60 items of the NEO-FFI-3 are presented on the second and third pages of the Item Booklet. The three validity check items and an area for recording item responses are presented on the third page. The demographic information and item responses are reproduced on the bottom sheet. The bottom sheet provides templates for scoring the domain scales of the NEO-FFI-3 and provides profile areas for converting raw scores to *T* scores and for graphing the individual's scores. *T* scores are based on age-specific data for male, female, and combined-gender norms. Your NEO

Summary provides a description of the individual's personality traits on the five factors that can be used to provide feedback. The NEO Style Graph Booklet summarizes implications of scores on pairs of factors.

Supplementary NEO Materials

NEO Software System

The NEO Software System offers a comprehensive system for administering, scoring, and interpreting the NEO Inventories (NEO-PI-3, NEO PI-R, and NEO-FFI-3). The Software System generates both detailed interpretive reports and client feedback reports. The structure of the comprehensive interpretive report is described in chapter 5 of this Professional Manual. The NEO Style Graph Booklet and the NEO Problems in Living Checklist, discussed later in this section, are included as options in the interpretive report.

Your NEO Summary

Your NEO Summary is a one-page, 8½" x 11" form designed to provide feedback to the individual in understandable terms regarding his or her scores on the five domains. It may be used to report the results of the five domain or factor scores from the NEO-PI-3, the NEO PI-R, or the NEO-FFI-3. The top portion contains an introductory statement explaining what the NEO Inventories do and do not measure. The remainder of the form presents five rows of descriptions, one for each domain, which characterize high ($T > 55$), average ($T = 45-55$), and low ($T < 45$) scorers within each domain. The test administrator should check off the appropriate boxes based on the individual's scores before providing it to the individual.

Your NEO Summary can be used with either Form S or Form R data. If feedback is based on observer ratings (Form R), *it is the professional's responsibility to ensure that informed consent for feedback has been obtained from both the rater and the target.*

NEO Style Graph Booklet

The NEO Style Graph Booklet is intended to assist in understanding the implications of specific pairs of factor scores, and may be used by professionals or included as part of feedback to respondents. The 10 graphs are based on the 10 combinations of five factors, taken two at a time (e.g., N and E, N and O). NEO Style Graph Booklets can be used with *T* scores from the NEO-PI-3, the NEO PI-R, or the NEO-FFI-3 and from Form S or Form R.

NEO Problems in Living Checklist

The NEO Problems in Living Checklist (NEO-PLC) is a tool that helps clinicians assess problematic behaviors

and symptoms associated with an individual's NEO-PI-3 profile. Problems described on the NEO-PLC were identified through content analyses and a review of relevant literature. The list presents potential problems that may or may not be relevant to a particular examinee for each NEO domain and facet. The checklist is designed to serve as a guide for a focused clinical interview. The full NEO-PI-3 (or NEO PI-R) must be administered before the NEO-PLC can be completed.

NEO-PI-3 and NEO-FFI-3 Licensing Agreements

Occasionally, professionals may wish to employ only a subset of the NEO-PI-3 scales or to present or score the scales in a manner other than the standard format. For example, researchers may have time to include only the specific scales that they hypothesize will be related to their criterion, or they may wish to administer NEO-PI-3 items as part of a preprinted assessment battery. Similarly, the need may arise for a test component that has not yet been published, such as a foreign language version of the instrument. In all such cases, the professional must obtain a formal, written licensing agreement from the publisher (PAR).

Unless there are compelling reasons, such ad hoc versions are discouraged. One of the great strengths of the instrument is its comprehensiveness, which makes systematic research possible. If some scales are omitted, important relations with other criteria may be missed. Further, nonstandard versions of the instrument may contain errors in the phrasing of items, the keying of scales, or the application of norms. The accuracy and validity of these licensed derivations remain the responsibility of the professional who holds the license.

Foreign Language Versions

The Five-Factor Model and the NEO Inventories have become the focus of personality research worldwide; the NEO-PI-3 has been used to rate adolescents age 12 to 17 years in 24 cultures. To date, the NEO Inventories have been translated into more than 50 languages and dialects. Table 2 lists 24 validated, published versions; professionals who wish to use these versions should contact the publisher. In addition, many other translations have been made, reviewed, and authorized; some of these have supporting data. Researchers interested in these unpublished

Table 2
Published Foreign Language Versions of the NEO Inventories

Language	Publisher	Web site
British English	The Test Agency, Hogrefe	http://www.hogrefe.co.uk
Brazilian Portuguese	Vetor Editora, São Paulo	http://www.vetoreditora.com.br
Bulgarian	Organizzazioni Speciali	
Croatian	Naklada Slap	http://www.nakladaslap.com
Czech	Hogrefe Testcentrum, Prague	http://www.testcentrum.com
Danish	Dansk Psykologisk Forlag	http://www.dpf.dk
Dutch	Hogrefe, Amsterdam	http://www.hogrefe.nl
Finnish	Hogrefe Psylogiforlaget	
French	Hogrefe, Paris	http://www.hogrefe.fr
French Canadian	Institute de Recherches Psychologiques, Montreal	http://www.i-r-p.ca
German	Hogrefe, Göttingen	http://www.testzentrale.de
Hebrew	BIP Institute of Psychology, Bnei Brak	
Japanese	Tokyo Shinri	
Korean	PSI Consulting, Seoul	
Lithuanian	Psichologija, Vilnius	
Norwegian	Gyldendal Akademisk	
Polish	Pracownia Testow Psychologicznych	http://www.practest.com.pl
Portuguese	CEGOC-TEA, Lisbon	http://www.cegoc.pt
Romanian	D&D Consultants Grup	http://www.testcentral.ro
Russian	Psychodiagnostica, Yaroslavl	
Slovak	Hogrefe Testcentrum	http://www.testcentrum.com
Spanish (Castilian)	TEA Ediciones, Madrid	http://www.teaediciones.com
Swedish	Psykologiförlagt, Stockholm	http://www.hogrefe.se
Turkish	Assess İnsan Gucu Degerlendirme ve danismanhk, Istanbul	

Note. A Spanish-language version suitable for use in the United States is published by PAR. For other published versions, see the table above. Contact PAR information on unpublished translations.

foreign language versions should contact the publisher's Customer Support department for licensing information.

We discourage multiple translations into the same language, although we understand that adaptations for particular dialects are often needed. Researchers who wish to prepare a new translation or modify an existing one must first obtain a license from the publisher. Information on the process of translating the NEO Inventories is presented in Appendix C of this Professional Manual.

Appropriate Populations

The NEO-PI-3 and NEO-FFI-3 may be appropriately used with individuals who are 12 years of age or older. Individuals who are suffering from disorders that affect their ability to complete self-report measures reliably and validly (e.g., acute psychosis, dementia) should not be asked to complete the NEO-PI-3 or NEO-FFI-3, but research has shown that individuals with a wide range of psychiatric disorders including, for example, schizophrenia, can provide valid data on these instruments (Costa & McCrae, 2009). The NEO Inventories can be administered orally to individuals with limited literacy or with visual problems.

Some respondents are likely to be unwilling or unable to give accurate self-descriptions for a variety of reasons. In these cases, psychologists should consider obtaining personality ratings from competent and unbiased informants using Form R.

The NEO-PI-3 does not provide all the information needed for a complete psychological assessment. Depending on the purpose of the assessment and the individual being assessed, life history interviews, mental status examinations, measures of cognitive ability, or medical laboratory tests may be needed to assess the individual or establish an appropriate diagnosis and treatment plan. The NEO-PI-3 is designed to be part of a comprehensive psychological assessment.

Professional Requirements

The NEO-PI-3 and NEO-FFI-3 are essentially self-administered instruments. Thus, the administration and scoring of these instruments can be performed by individuals who do not have formal training in clinical psychology, personality, or related fields. The administration and scoring procedures detailed in this Professional Manual should be carefully studied by the examiner.

In keeping with the *Standards for Educational and Psychological Testing* (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 1999), interpretation of the NEO-PI-3 and NEO-FFI-3 requires professional training in psychological testing and measurement (e.g., reliability and validity, use of norms), in addition to familiarity with the materials and procedures presented in this Professional Manual. The utility of the NEO-PI-3 and NEO-FFI-3 is clearly related to the professional's background and knowledge.

Administration and Scoring of the NEO-PI-3

Administration

The NEO-PI-3 and NEO PI-R may be administered individually or in groups. The testing environment should be comfortable and free of distractions and provide adequate lighting. A pencil and a flat surface, such as a desk or clipboard on which the respondent can write, are also needed for administration. If the respondent uses eyeglasses, the examiner should be sure that the respondent uses them during testing. The examiner should engage the respondent in the task of completing the test to reduce the possibility of response sets or random responding to the items.

Directions

Provide each respondent with the appropriate NEO-PI-3 (or NEO PI-R) Item Booklet, a Hand-Scorable (HS) or Scannable (SS) Answer Sheet, and a pencil. A No. 2 lead pencil is required for completing the SS Answer Sheet. Depending on which Answer Sheet has been selected, tell the respondent to read the instructions for completing the NEO-PI-3 on the first or second page of the Item Booklet. Tell the respondent to provide the identifying information requested on the Answer Sheet. If an identification number is to be used, provide the respondent with the number so he or she can enter it, or fill it in yourself. If Form R is being used, the respondent should provide the initials of the person being rated. *It is the examiner's responsibility to ensure that the person being rated can be correctly identified by these initials, the identification number, and/or the name of the rater.* The examiner is also responsible for ensuring that all identifying information is properly marked on the Scannable Answer Sheet.

After the respondent has provided the demographic information and read the instructions for completing the NEO-PI-3, ask if there are any questions. When all questions have been answered to the respondent's satisfaction, tell the respondent to turn to the third page and begin. There is no time limit for completing the NEO-PI-3. Most respondents require 30 to 40 minutes to complete the measure, but older respondents and those with limited reading skills may take longer.

Alternative Administration Procedures and Accommodations

There are circumstances that dictate that the standard administration procedures be modified. In cases where the anonymity of the respondent must be guaranteed, the instructions for providing demographic information may be modified. Similarly, the presence of an examiner, although desirable for answering questions and observing test-taking behavior, is not required. The NEO Inventories have been used frequently in projects for which respondents completed the instrument at home. In settings where no examiner is present, the professional is responsible for protecting the integrity of the test by ensuring that all materials are returned. Finally, administration and recording procedures also can be modified to enable individuals with poor reading skills or who are visually impaired to complete the NEO-PI-3. In these cases, the examiner should read the items aloud and mark the examinee's responses on the Answer Sheet for the respondent (cf. Costa et al., 1986).

Although the items in the NEO-PI-3 and NEO-FFI-3 were chosen to be understandable to a wide range of adolescents and adults, occasionally respondents do not understand an item. In such cases, the respondent can be instructed to leave the item blank, and it will be treated as missing data. Alternatively, it is permissible for the test administrator to provide an explanation of the word or item. Table 3 provides suggested language for selected items. These paraphrases were reviewed by middle school teachers; however, they may be helpful for respondents of any age who have difficulty with the meanings of these items.

Scoring

Missing Responses

Once the measure has been completed, examine the Answer Sheet to be sure that a response has been given for each item. If unanswered items are found, the respondent should be asked to complete them. If a respondent is unsure of the meaning of an item or is unsure of how to respond, he or she may be told to use the *neutral* response

Table 3
Glossary of Less Familiar Words Used in the NEO-PI-3

The following is a list of words used in the NEO-PI-3 that some middle school-aged children did not understand. These definitions may be given to respondents of any age who express difficulty understanding the meaning of the item.

Word	NEO-PI-3 Item No.	Glossary definition
assertive	12	Speaking out and claiming my rights
seldom	21, 71, 103, 121, 163, 166	Not very often
impulses	21	Strong desires for things that may not be good for me, like candy
well-intentioned	34	Well-meaning; trying to be nice
manipulate	39	Trick
philosophical (arguments)	53	Talk about ideas that are important but hard to understand, like the meaning of life
hasty (decisions)	60	Quick choices made without thinking things through
methodical	70	In an systematic, organized way
evoke	73	Produce or create
calculating	74	Secretly planning or scheming to get what I want
spur (of the moment)	90	Immediately, without giving any thought
speculating	113	Thinking about possible answers
broad-minded	118	Respecting beliefs that differ from mine
hypocrite	129	Person who says one thing but believes another; a phony
egotistical	134	Concerned only with myself
ambitious	140	Wanting very much to succeed, to get ahead
happy-go-lucky	147	Carefree, untroubled, not serious
temperamental	156	Easily angered or offended
ethical (principles)	165	Beliefs about what I ought to do
theoretical (matters)	173	Ideas that are hard to understand and not part of real life
compulsive	190	Feel strongly that I have to do it
empathize	193	Understand how others feel
charitable	194	Generous, freely giving gifts or money
intellectual (curiosity/interests)	203, 233	Interest in ideas and in studying things
optimist	207	A person who thinks everything will turn out well
controversial (speakers)	208	People with ideas that some people strongly disagree with
conscientiously	225	Completely and carefully
merciful	239	Not strict, forgiving

Note. The authors thank Lori Lupolt, Faithe Bastian, Leola Gaugler, Glenda Nichols, Kelly King, and Karen Hoke of Selinsgrove Area Middle School, Selinsgrove, PA, for their advice on the phrasing of entries in this glossary.

option. It is also appropriate to explain the meaning of words the respondent does not understand (see the previous section).

If the respondent has not provided a response to every item and the respondent is no longer available to the examiner, the examiner must determine whether the data may be validly scored and interpreted. The NEO-PI-3 and NEO PI-R should not be scored if 41 or more responses are missing. If fewer than 41 responses are missing, the missing items should be scored as if the *neutral* response option was selected. In research applications in which large samples are used, the group mean for an item can be

substituted for the missing response. Individual facet scores should be interpreted with caution when more than three responses are missing from that facet.

Validity Checks

Items A, B, and C, which are presented on the Answer Sheet, provide simple validity checks to help ensure that the respondent has completely and accurately completed the NEO-PI-3. Item A asks if the respondent has responded to the items in an honest and accurate manner. Approximately 99% of all volunteer participants responded *agree* or *strongly agree* to Item A. Respondents who disagree may do so because they have not been fully candid,

because they have responded carelessly, or because they have confused the response categories. Whatever the reason, a *disagree* or *strongly disagree* response to Item A normally invalidates formal scoring of the NEO-PI-3. Clinicians or other users concerned with interpreting individual profiles may wish to discuss the reason for a negative response to this item with the respondent and, in some cases, may determine that the data are valid despite the response.

Items B and C ask if the respondent has answered all the items (Item B) and has marked his or her responses in the correct spaces (Item C), questions that are chiefly intended as prompts to the respondent to complete missing items and to double-check the location of his or her answers. If the respondent answered *No* to Item B, the examiner may wish to explore the respondent's reason for the missing responses and will need to attend closely to the instructions for treating missing items. The response of *No* to Item C may indicate careless responding and normally invalidates formal scoring of the NEO-PI-3. The professional may wish to discuss the reason for this response with the respondent to determine whether or not the data are valid.

Acquiescence

All NEO-PI-3 scales are roughly balanced to control for the effects of acquiescence. This means that a respondent who endorses an excessive number of *agree* or *strongly agree* responses will tend to receive average scores instead of extreme—and perhaps extremely misleading—scores. To screen for acquiescence, count the number of *agree* and *strongly agree* responses made across all items. In a large volunteer sample (Costa, McCrae, & Dye, 1991), 99% of respondents agreed with fewer than 150 items. Thus, if the respondent has 150 or more *agree* and *strongly agree* responses, the NEO-PI-3 should be interpreted with caution because a strong acquiescence bias may have influenced the results.

Nay-Saying

Nay-saying on the NEO-PI-3 is indicated by a paucity of *agree* or *strongly agree* responses. To screen for nay-saying, count the number of *agree* and *strongly agree* responses made across all items. Approximately 99% of a large volunteer sample (Costa, McCrae, et al., 1991) agreed with more than 50 items. If the respondent endorsed 50 or fewer items as *agree* or *strongly agree*, the NEO-PI-3 should be interpreted with caution.

Random Responding

Occasionally, respondents are uncooperative and complete the NEO-PI-3 in a careless or random fashion. This is most likely to occur when the test is group-administered

and the examiner has not provided sufficient explanation of the purpose for testing. Unmotivated test-takers often respond in a haphazard manner simply to complete the assigned task quickly. Ideally, the examiner can avoid this problem by ensuring that the NEO-PI-3 is presented in a way that will engage respondents.

One common form of random responding can be evaluated by visually inspecting the Answer Sheet to determine whether the same response option has been used over a long series of items. Based on the results of item response patterns in a volunteer sample (Costa & McCrae, 2008b), endorsements of *strongly disagree* to more than six consecutive items, *disagree* to more than nine consecutive items, *neutral* to more than 10 consecutive items, *agree* to more than 14 consecutive items, or *strongly agree* to more than nine consecutive items invalidate formal scoring and interpretation of the NEO-PI-3.

All these validity checks are included in the NEO Software System interpretive report.

Calculating Scores

Hand-Scorable (HS) Answer Sheet

Tear off the perforated stub at the top of the Answer Sheet and remove the top page. Locate the first row of items (i.e., Items 1, 31, 61, 91, 121, 151, 181, and 211) and sum the values of the circled responses to these eight items. Enter this sum in the space on the right labeled *N1*. This number is the raw score for facet N1. Use an analogous procedure to calculate the remaining facet raw scores.

After all facet scores have been calculated, add the scores for facets N1, N2, N3, N4, N5, and N6. Enter this sum in the box labeled *N* within the Domain Raw Score table at the bottom of the answer sheet. This number is the raw score for the N domain. Use an analogous procedure to calculate the remaining domain raw scores.

Scannable (SS) Answer Sheet

The Scannable (SS) Answer Sheet is scored and interpreted through the Professional Report Service available through the publisher. Return completed SS Answer Sheets to the publisher along with a completed control sheet for the mail-in Professional Report Service for scoring and interpretation.

Researchers and other professionals who collect large quantities of data occasionally need to obtain NEO-PI-3 scores but do not need the interpretive reports which accompany the results of the Professional Report Service. In these cases, special arrangements can be made with the publisher. Professionals interested in this service should contact PAR *before* ordering NEO-PI-3 testing materials.

Profiling Scores

When the Hand Scorable (HS) Answer Sheet is used, NEO-PI-3 results may be presented on a Profile Form to facilitate raw score to standard score conversions. Profile Forms provide *T* scores based on different normative samples. Separate Profile Forms are available for adults and adolescents and for Form S and Form R. The Adult Profile Forms are appropriate for use with individuals age 21 years and older. The Adolescent Profile Forms are appropriate for use with individuals age 12 to 20 years. Profiles are plotted separately for men and women, unless the combined norms are being used. When Form R scores are plotted on the Form R Profile Form, the *sex of the individual rated* determines which form to use—the sex of the rater is irrelevant.

If the 1992 NEO PI-R is administered, NEO PI-R Profile Forms must be used; NEO-PI-3 Profile Forms are based on different norms.

Form R ratings for an individual from multiple raters may be averaged to increase the accuracy of the ratings; these averages may be plotted on the Profile Form. Alternatively, multiple observer ratings can be plotted on one Form R Profile Form for comparison. However, when self-reports and observer ratings are compared, the associated scores cannot be plotted directly on the same form because different norms are used in determining the *T* scores. In these cases, the Form S Profile Form can be used to determine *T* scores for the self-report. These *T* scores can then be plotted on the Form R Profile Form (according to the *T* score values, not raw score values), along with the Form R ratings (see Case D in chapter 5 for an example).

Directions

Select the appropriate Profile Form based on the NEO-PI-3 form administered and the relevant comparison group. Complete the demographic information at the top of the Profile Form. Locate the raw score for the N domain at the bottom of the Answer Sheet and transcribe this score to the space at the bottom of the Profile Form labeled *N*. Use an analogous procedure to transcribe the remaining domain raw scores.

Similarly, locate the raw score for the N1 facet on the right side of the Answer Sheet and transcribe this score to the space at the bottom of the Profile Form labeled *N1*. Use an analogous procedure to transcribe the remaining facet raw scores. Add the transcribed facet scores for N1, N2, N3, N4, N5, and N6 and enter this number in the space labeled *Total (N)* at the bottom of the Profile Form. Compare this number to the N domain raw score entered on the left-hand side of the Profile Form. If the numbers do

not match, an error in addition or transcription has been made. Locate and correct the error. Use an analogous procedure to check the accuracy of the remaining facet scores.

In the column for the N domain, locate the number that corresponds to the N domain raw score and mark it with an X. Use a similar procedure to mark the remaining domain and facet scores on the Profile Form. The *T* scores corresponding to each mark are then read off the scale in the column marked *T* at the left or right side of the profile. Write the *T* score for each scale in the space provided at the top of the profile below each domain or facet name. These *T* scores are used in Your NEO Summary, the NEO Style Graph Booklet, and the NEO Problems in Living Checklist. After all scores have been marked, connect the Xs of the domain scales and the Xs of the facet scales within each domain with a line to produce a graph of the individual's scores. See Figure 8 in chapter 5 for an example of a completed NEO-PI-3 Profile Form.

Calculating and Using Factor Scores

The five domains of the NEO-PI-3 correspond to the five factors of personality that have been found in the analysis of many different instruments, including factor analyses of the 30 NEO-PI-3 facets. Chapter 8 of this Professional Manual, *Validation of the Inventories and Materials*, presents the results of a varimax-rotated principal components analysis of Form S NEO PI-R data from its normative sample and shows that the domain scales are strongly correlated with their associated factors. However, from a psychometric point of view, factor scores themselves are slightly better than domain scores. Because factor scores combine information from each of the 30 facets to estimate each of the five factors, they benefit from the secondary factor loadings. Factor scores are more nearly orthogonal and tend to have somewhat higher validities against external criteria (McCrae & Costa, 1989c; McCrae et al., 2008).

For these reasons, computerized versions of the NEO-PI-3 (i.e., the Professional Report Service and the NEO Software System) compute factor scores and use them in generating the interpretive profile. Researchers who hand-score the NEO-PI-3 can use the formulas in Table 4 to calculate factor scores. These formulas must be applied to *T* scores, not raw scores, and yield factor scores expressed as *T* scores. The factor score weights are based on analyses of the Form S NEO PI-R sample, which provides the gold standard for the structure of NEO Inventories. Because, as Table 6 in chapter 7 shows, the factor structure of the NEO-PI-3 is almost identical to that of the NEO PI-R, it is appropriate to use the weights in Table 4 to score factors for the NEO-PI-3 (once *T* scores based on

Table 4
Formulas for Calculating Form S and Form R Factor Scores from NEO PI-R T Scores

Form S factor scores:											
N Factor =	(.26N1	+	.18N2	+	.23N3	+	.22N4	+	.11N5	+	.18N6
	+ .01E1	-	.06E2	-	.07E3	+	.08E4	-	.02E5	+	.02E6
	+ .02O1	+	.09O2	+	.16O3	-	.06O4	-	.02O5	-	.06O6
	- .09A1	+	.05A2	+	.05A3	-	.02A4	+	.07A5	+	.05A6
	- .03C1	+	.10C2	+	.05C3	+	.09C4	+	.01C5	+	.02C6)
	- 31.00										
E Factor =	(.02N1	+	.00N2	-	.02N3	-	.04N4	+	.16N5	-	.01N6
	+ .21E1	+	.24E2	+	.10E3	+	.15E4	+	.21E5	+	.24E6
	- .01O1	-	.12O2	+	.07O3	-	.01O4	-	.14O5	-	.05O6
	+ .05A1	-	.05A2	+	.19A3	-	.03A4	-	.01A5	+	.08A6
	- .01C1	+	.01C2	-	.07C3	+	.01C4	+	.02C5	-	.14C6)
	- 2.50										
O Factor =	(.00N1	+	.00N2	+	.03N3	+	.00N4	-	.06N5	-	.01N6
	- .02E1	-	.09E2	+	.02E3	-	.02E4	-	.06E5	-	.03E6
	+ .23O1	+	.34O2	+	.17O3	+	.22O4	+	.35O5	+	.21O6
	+ .05A1	+	.00A2	-	.09A3	+	.03A4	-	.04A5	+	.03A6
	+ .04C1	-	.09C2	+	.03C3	+	.04C4	-	.05C5	+	.04C6)
	- 13.50										
A Factor =	(.03N1	-	.12N2	+	.03N3	+	.05N4	-	.04N5	+	.05N6
	+ .12E1	+	.02E2	-	.12E3	-	.09E4	-	.11E5	+	.03E6
	- .01O1	+	.08O2	+	.02O3	+	.02O4	-	.02O5	-	.01O6
	+ .16A1	+	.20A2	+	.16A3	+	.23A4	+	.19A5	+	.20A6
	- .02C1	-	.03C2	+	.06C3	-	.06C4	-	.02C5	+	.04C6)
	- 2.00										
C Factor =	(.09N1	+	.09N2	+	.04N3	+	.07N4	-	.05N5	-	.02N6
	- .03E1	-	.09E2	+	.05E3	+	.13E4	-	.05E5	-	.02E6
	- .08O1	+	.08O2	+	.08O3	-	.05O4	+	.05O5	-	.07O6
	- .08A1	+	.07A2	+	.03A3	-	.04A4	-	.01A5	-	.03A6
	+ .16C1	+	.24C2	+	.21C3	+	.25C4	+	.21C5	+	.18C6)
	- 20.50										
Form R factor scores:											
N Factor =	(((((.26N1	+	.18N2	+	.23N3	+	.22N4	+	.11N5	+	.18N6
	+ .01E1	-	.06E2	-	.07E3	+	.08E4	-	.02E5	+	.02E6
	+ .02O1	+	.09O2	+	.16O3	-	.06O4	-	.02O5	-	.06O6
	- .09A1	+	.05A2	+	.05A3	-	.02A4	+	.07A5	+	.05A6
	- .03C1	+	.10C2	+	.05C3	+	.09C4	+	.01C5	+	.02C6)
	- 31.00)	- 50.00)	*	1.031)	+	50.00					
E Factor =	(((((.02N1	+	.00N2	-	.02N3	-	.04N4	+	.16N5	-	.01N6
	+ .21E1	+	.24E2	+	.10E3	+	.15E4	+	.21E5	+	.24E6
	- .01O1	-	.12O2	+	.07O3	-	.01O4	-	.14O5	-	.05O6
	+ .05A1	-	.05A2	+	.19A3	-	.03A4	-	.01A5	+	.08A6
	- .01C1	+	.01C2	-	.07C3	+	.01C4	+	.02C5	-	.14C6)
	- 2.50)	- 50.00)	*	1.068)	+	50.00					
O Factor =	(((((.00N1	+	.00N2	+	.03N3	+	.00N4	-	.06N5	-	.01N6
	- .02E1	-	.09E2	+	.02E3	-	.02E4	-	.06E5	-	.03E6
	+ .23O1	+	.34O2	+	.17O3	+	.22O4	+	.35O5	+	.21O6
	+ .05A1	+	.00A2	-	.09A3	+	.03A4	-	.04A5	+	.03A6
	+ .04C1	-	.09C2	+	.03C3	+	.04C4	-	.05C5	+	.04C6)
	- 13.50)	- 50.00)	*	0.975)	+	50.00					
A Factor =	(((((.03N1	-	.12N2	+	.03N3	+	.05N4	-	.04N5	+	.05N6
	+ .12E1	+	.02E2	-	.12E3	-	.09E4	-	.11E5	+	.03E6
	- .01O1	+	.08O2	+	.02O3	+	.02O4	-	.02O5	-	.01O6
	+ .16A1	+	.20A2	+	.16A3	+	.23A4	+	.19A5	+	.20A6
	- .02C1	-	.03C2	+	.06C3	-	.06C4	-	.02C5	+	.04C6)
	- 2.00)	- 50.00)	*	0.867)	+	50.00					
C Factor =	(((((.09N1	+	.09N2	+	.04N3	+	.07N4	-	.05N5	-	.02N6
	- .03E1	-	.09E2	+	.05E3	+	.13E4	-	.05E5	-	.02E6
	- .08O1	+	.08O2	+	.08O3	-	.05O4	+	.05O5	-	.07O6
	- .08A1	+	.07A2	+	.03A3	-	.04A4	-	.01A5	-	.03A6
	+ .16C1	+	.24C2	+	.21C3	+	.25C4	+	.21C5	+	.18C6)
	- 20.50)	- 50.00)	*	0.948)	+	50.00					

Note. From the *Revised NEO Personality Inventory (NEO PI-R) and NEO Five-Factor Inventory (NEO-FFI) Professional Manual* (p. 8), by P. T. Costa, Jr. & R. R. McCrae, 1992, Odessa, FL: PAR. Copyright 1992 by PAR. Reprinted with permission.

NEO-PI-3 norms have been calculated). Because very similar factors are found in analyses of Form R data, the same weights are used to calculate factor scores from Form R *T* scores. However, variances are somewhat different in Form R scales, especially for Agreeableness. The formulas presented in Table 4 for calculating Form R factor scores correct for these differences.

In some cases with unusual configurations of facet scores, the domain and factor *T* scores differ appreciably. However, the domain scale scores are usually a good approximation to factor scores, and it is probably not worth the effort to apply these formulas by hand to individual cases. Researchers who wish for maximum precision and who are concerned about the orthogonality of scores, however, may wish to apply these formulas to their data.

Providing Feedback to the Respondent

The NEO-PI-3 is intended as a tool for professionals trained in the use and interpretation of psychological tests. Generally, it would not be appropriate to provide respondents with copies of their test scores, profiles, or interpretive reports. Exceptions may include clinical practice settings where the clinician believes that the client is capable of understanding and benefiting from detailed feedback on personality scores and educational settings where the NEO-PI-3 is part of a course on psychological testing. But respondents often appreciate feedback, and experience with the NEO PI-R and other instruments has shown that most individuals can benefit from suitably phrased descriptions of their personality. Your NEO Summary and the NEO Style Graph Booklet were designed to serve this purpose.

Your NEO Summary

Examine the individual's *T* score for the *N* domain or factor. *T* scores of 56 or higher are considered high, *T* scores ranging from 45 to 55 are considered average, and *T* scores of 44 or lower are considered low. Locate the first row of descriptions on the Your NEO Summary form. This row contains descriptions for the Neuroticism domain. If the individual's score is high, place a check mark in the left-hand box. If the individual's score is average, place a check mark in the middle box. If the individual's score is low, place a check mark in the right-hand box. Use an analogous procedure to complete the remainder of the form. The second through fifth rows of descriptions correspond to the *E*, *O*, *A*, and *C* domains, respectively.

Figure 1 presents an example of the feedback that would be provided to an individual with *T* scores of 48, 58, 33, 50, and 56 on the *N*, *E*, *O*, *A*, and *C* domains,

respectively. Because the word "high" may be construed as "good" by respondents, Your NEO Summary does not characterize scores as high or low.

Your NEO Summary may be useful in educational settings and in research, where it may be included as part of the subject's debriefing. In clinical settings, it may serve as a source of insight as part of what McReynolds (1989) called "client-centered assessment." For clients who are administered the NEO-PI-3 via the computerized administration option in the NEO Software System, an optional Client Report gives a more detailed lay summary of results, including facet scores. This option is available only for Form S, where feedback is based on the client's own responses.

For some clients, feedback may be inappropriate. It is the professional's responsibility to determine whether simple or detailed feedback is appropriate and to provide the individual with an opportunity to discuss the results if he or she desires it.

NEO Style Graph Booklet

For some purposes, professionals may wish to provide more extensive feedback that takes into account configurations of factors; the NEO Style Graph Booklet provides such feedback. At the top of each graph, enter the appropriate domain or factor *T* score for the vertical axis and the appropriate *T* score for the horizontal axis. Plot these values on the appropriate axis with Xs and draw vertical and horizontal lines to find the point of their intersection. Mark this point with an X. This point identifies the style of the respondent, and the description associated with that quadrant of the graph characterizes his or her style.

Figure 2 shows an example for the Style of Interests. The individual whose scores are plotted has a low *E* score ($T = 44$) and a high *O* score ($T = 56$), and these scores intersect in the lower right quadrant, labeled *Introspectors*. The text in this quadrant describes the interest style of this individual. Note that some respondents will have near-average scores on both plotted factors; their scores will intersect in the shaded central region. These respondents (about 10% of individuals) are not strongly characterized by any of the four styles described but share some features of all of them.

For some applications, only certain styles may be of interest. For example, in counseling students, the Style of Learning may be the most relevant graph; in dealing with married couples, the Style of Interactions, which is defined by the two interpersonal dimensions of *E* and *A*, may be most revealing.

Your NEOTM Summary

Paul T. Costa, Jr., PhD and Robert R. McCrae, PhD

The NEO Inventory measures five broad domains, or dimensions, of personality. Responses describing your thoughts, feelings, and goals can be compared with those describing others' to portray your personality.

This summary is intended to give you a general idea of what your personality is like. It is not a detailed report. If you completed the inventory again, or if someone else described you, you might score somewhat differently.

The NEO Inventory measures differences among people in general. It is not a test of intelligence or ability, and it is not intended to diagnose problems of mental health or adjustment. It does, however, give you some idea of what makes you unique in your ways of thinking, feeling, and interacting with others.

Compared with the responses of other people, your responses suggest that you can be described as:

- | | | |
|--|---|--|
| ☐ Sensitive, emotional, and prone to experience feelings that are upsetting. | ☒ Generally calm and able to deal with stress, but you sometimes experience feelings of guilt, anger, or sadness. | ☐ Secure, hardy, and generally relaxed, even under stressful conditions. |
| ☒ Extraverted, outgoing, active, and high-spirited. You prefer to be around people most of the time. | ☐ Moderate in activity and enthusiasm. You enjoy the company of others, but you also value privacy. | ☐ Introverted, reserved, and serious. You prefer to be alone or with a few close friends. |
| ☐ Open to new experiences. You have broad interests and are very imaginative. | ☐ Practical, but willing to consider new ways of doing things. You seek a balance between the old and the new. | ☒ Down-to-earth, practical, traditional, and pretty much set in your ways. |
| ☐ Compassionate, good-natured, and eager to cooperate and avoid conflict. | ☒ Generally warm, trusting, and agreeable, but you can sometimes be stubborn and competitive. | ☐ Hardheaded, skeptical, proud, and competitive. You tend to express your anger directly. |
| ☒ Conscientious and well-organized. You have high standards and always strive to achieve your goals. | ☐ Dependable and moderately well-organized. You generally have clear goals, but are able to set your work aside. | ☐ Easygoing, not very well-organized, and sometimes careless. You prefer not to make plans. |

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Figure 1. Example of a completed Your NEO Summary for a respondent with *T* scores of 48, 58, 33, 50, and 56 on the N, E, O, A, and C domains, respectively.

Style of Interests

Vertical Axis: Extraversion

$T = 44$

Horizontal Axis: Openness

$T = 56$

E+O-

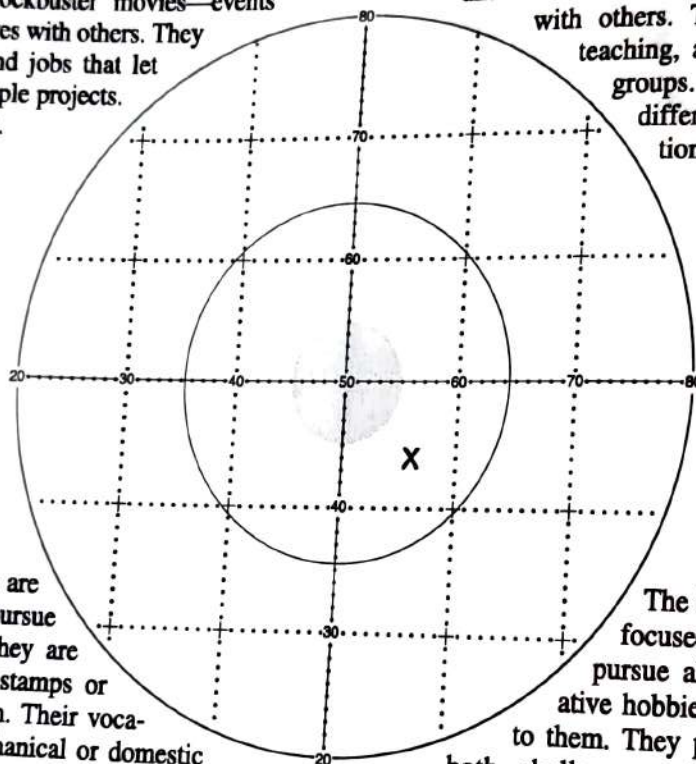
Mainstream Consumers

The interests of these individuals reflect popular favorites: parties, sports, shopping, blockbuster movies—events where they can enjoy themselves with others. They are attracted to businesses and jobs that let them work with others on simple projects. Possible vocation: Salesperson.

E+O+

Creative Interactors

The interests of these individuals revolve around the new and different, and they like to share their discoveries with others. They enjoy public speaking and teaching, and they fit in well in discussion groups. They enjoy meeting people from different backgrounds. Possible vocation: Anthropologist.



E-O-

Homebodies

The interests of these individuals are focused on activities they can pursue alone or with a small group. They are unadventurous and may collect stamps or coins, watch television, or garden. Their vocational interests may include mechanical or domestic work. Possible vocation: Bookkeeper.

E-O+

Introspectors

The interests of these individuals are focused on ideas and activities they can pursue alone. Reading, writing, or creative hobbies (e.g., painting, music) appeal to them. They prefer occupations that provide both challenge and privacy. Possible vocation: Naturalist.

Figure 2. Example of a completed Style of Interests graph from the NEO Style Graph Booklet.

4

Administration and Scoring of the NEO-FFI-3

The NEO-FFI-3 is a 60-item version of the NEO-PI-3 that provides a brief, comprehensive measure of the five domains of personality. It consists of five 12-item scales that measure each domain. Information on specific facets within each domain is not provided, and the shortened scales are somewhat less reliable and valid than the full NEO-PI-3 domain scales. Separate versions are available for Form S and Form R and for adults and adolescents.

Administration

The NEO-FFI-3 may be administered individually or in groups. The testing environment should be comfortable and free of distractions and provide adequate lighting. A pencil and a flat surface, such as a desk or clipboard on which the respondent can write, are also needed for administration. If the respondent uses eyeglasses, the examiner should be sure that they are being used during testing. The examiner should engage the respondent in the task of completing the test to reduce the possibility of response sets or random responding to the items.

Directions

Provide each respondent with a NEO-FFI-3 Item Booklet and a pencil. Tell the respondent to read the instructions for completing the NEO-FFI-3, which are on the first page. After the instructions have been read and the respondent understands the nature of the task, tell him or her to turn to the second page of the Item Booklet and to provide the identifying information requested at the top of that page. If an identification number is to be used, provide the respondent with the number so he or she can enter it, or fill it in yourself.

If Form R is being used, the respondent should provide the name (or initials) of the person being rated. *It is the examiner's responsibility to ensure that the person being rated can be correctly identified by name, initials, ID number, and/or the name of the rater.* The examiner is also responsible for ensuring that all identifying information is properly marked on the Item Booklet. Note that responses to the NEO-FFI-3 are made across the rows of the response grid.

There is no time limit for the NEO-FFI-3. Most respondents require 5 to 10 minutes to complete the measure, but older respondents and those with limited reading skills may take longer. The NEO-PI-3 alternative administration procedures previously described may also be used with the NEO-FFI-3.

Scoring

Missing Responses

Once the measure has been completed, examine the response area of the Item Booklet to be sure that a response has been given for each item. If unanswered items are found, the respondent should be asked to complete them. If a respondent is unsure of the meaning of an item or is unsure of how to respond, he or she may be told to use the *neutral* response option. It is also appropriate to explain the meaning of words the respondent does not understand (see chapter 3 of this Professional Manual for more information).

If the respondent did not provide a response to every item and the respondent is no longer available to the examiner, the examiner must determine whether the data may be validly scored and interpreted. If 10 or more items have been left blank, the test is considered invalid and should not be formally scored. If nine or fewer items have been left blank, the blank items should be scored as if the *neutral* response option was selected. Any domain for which there are more than four missing responses should be interpreted with caution.

Validity Checks

At the bottom of the third page of the NEO-FFI-3 Item Booklets, three yes-or-no items ask the respondent if he or she has (a) responded to all of the statements, (b) entered responses across the rows, and (c) responded accurately and honestly. In most cases, if the respondent indicated that responses were not entered in the correct boxes or were not answered honestly and accurately, the test should not be scored.

Calculating Domain Scores

Open the NEO-FFI-3 Item Booklet, tear off the perforated stubs on the left- and right-hand sides of the booklet, and remove the top page. Locate the first column of items (i.e., Items 1, 6, 11, 16, 21, 26, 31, 36, 41, 46, 51, and 56) in the answer grid and sum the values of the marked responses to these 12 items. Enter this sum in the space labeled *N*, which is located beneath the column. This number is the raw score for the *N* domain. Use an analogous procedure to calculate the remaining domain raw scores.

Profiling Scores

Profile areas for males or females (depending on which form is being used) and for a combined-gender group are provided on the left-hand side of the bottom sheet. Locate the column labeled *N* on the appropriate profile. Within this column, mark an X on the number that corresponds to the respondent's *N* domain raw score. Use an analogous procedure to mark the remaining domain raw scores. The *T* scores corresponding to each mark are then read off the scale in the column marked *T* at the left side of the profile. After all the respondent's scores have been marked, connect the Xs with a line to produce a graph of the respondent's NEO-FFI-3 scores.

Providing Feedback to the Respondent

After scoring and profiling the results of the NEO-FFI-3, the professional may wish to provide feedback to the

individual about his or her scores. Your NEO Summary was designed to serve this purpose. Examine the respondent's *T* score for the *N* domain. *T* scores of 56 or higher are considered high, *T* scores ranging from 45 to 55 are considered average, and *T* scores of 44 or lower are considered low. Locate the first row of descriptions on the Your NEO Summary form. This row contains descriptions for the Neuroticism domain. If the individual's score is high, place a check mark in the left-hand box. If the individual's score is average, place a check mark in the middle box. If the individual's score is low, place a check mark in the right-hand box. Use an analogous procedure to complete the remainder of the form. The second through fifth rows of descriptions correspond to the *E*, *O*, *A*, and *C* domains, respectively.

For certain purposes, professionals may wish to provide more extensive feedback that takes into account configurations of factors; the NEO Style Graph Booklet provides such feedback. At the top of each graph, enter the appropriate domain or factor *T* score for the vertical axis and the appropriate *T* score for the horizontal axis. Plot these values on the appropriate axis with Xs and draw vertical and horizontal lines to find the point of their intersection. Mark this point with an X. This point identifies the style of the respondent, and the description associated with that quadrant of the graph characterizes his or her style.

Conceptualization and Interpretation

Although some of the items have been replaced, the NEO-PI-3 measures the same traits as the NEO PI-R, and the conceptualization of those traits remains the same. Research since the publication of the NEO PI-R has confirmed the original interpretation of the scales and has shown that the trait concepts of the NEO Inventories are applicable to adolescents as well as adults and to individuals from cultures around the world. Professionals familiar with the NEO PI-R can transfer their knowledge and experience directly to the use of the NEO-PI-3.

In order to interpret NEO-PI-3 results, the professional must be familiar with the basics of psychological testing, know what the scales measure and their implications for the psychological functioning of the individual, and be able to integrate the scale score information into a meaningful profile. This chapter provides the basic information on the constructs that the NEO-PI-3 measures and suggests guidelines for interpreting profiles. Professionals may benefit from reading the computer-generated interpretive reports; some examples of case studies are provided later in this chapter. In addition, we recommend an interactive approach to learning the fine points of interpretation in which the research literature is read in conjunction with the actual profiles the professional needs to interpret. For example, the concept of Openness to Experience (O)—frequently referred to as simply Openness—is treated extensively in chapters and articles (e.g., McCrae, 1994b; McCrae & Costa, 1997a; McCrae & Sutin, 2009). It may be most useful to read these chapters when dealing with an extremely high or low scorer on the dimension of Openness. In this way, the respondent can illustrate the literature, and the literature can illuminate the case.

The Meaning of Scale Scores

The scales of the NEO-PI-3 and the NEO-FFI-3 measure traits that approximate normal, bell-shaped distributions. Most individuals will obtain scores near the average for the scale, with a small percentage scoring at either end. It is necessary to keep this distribution of scores in mind when interpreting the meaning of any individual's scores.

Scales are most conveniently explained by describing characteristics of extremely high or extremely low scorers. However, few individuals will obtain these scores or show all the characteristics described. Instead, individual scores will usually represent degrees of the personality trait; more extreme scorers have a higher probability of showing the distinctive features.

Characteristics are compared across people, rather than within the individual. Thus, a person who scores at the 75th percentile on the N3: Depression scale and at the 25th percentile on the E6: Positive Emotions scale is more likely than most other people to feel depressed and less likely than most other people to feel happy. But because happiness is much more common than depression, such an individual is still likely to be happy more often than he or she is depressed.

Professionals using the NEO-PI-3 should avoid thinking in terms of either types or categories when interpreting scores (McCrae, Terracciano, Costa, & Ozer, 2006). Although it is convenient to speak of "introverts" and "extraverts," the NEO-PI-3 Extraversion (E) scale represents a continuous dimension. Most individuals are best described as "ambiverts;" that is, they show a combination of introverted and extraverted tendencies. The same principle applies to all other domains and facets.

In designing the Profile Forms and Interpretive Reports for the NEO-PI-3, we have found it useful to summarize results in terms of five levels: Very Low, Low, Average, High, and Very High (just as test takers are asked to respond along a five-point scale from *strongly disagree* to *strongly agree*). Although the percentages vary somewhat with the exact shape of the distributions (the NEO Inventories do not use normalized *T* scores), of all individuals administered the NEO-PI-3, approximately 38% score in the Average range ($T = 45$ to 55), 24% score in the High range ($T = 56$ to 65), 24% score in the Low range ($T = 35$ to 44), 7% score in the Very High range ($T = 66$ and higher), and 7% score in the Very Low range ($T = 34$ and lower). This system allows for more fine-grained distinctions than simple dichotomies, but it is not needlessly complex.

Inventories like the Minnesota Multiphasic Personality Inventory (MMPI, Hathaway & McKinley, 1983) are often interpreted in terms of diagnostic categories. If a scale exceeds a *T* score of 70, for example, it may be regarded as an indication of psychopathology; scores below this cutoff are considered normal and sometimes given little attention. In conceptualizing the personality traits measured by the NEO-PI-3, a different approach should be used. No single cutoff point separates those who "have" a trait from those who do not, and being low or average on a scale can be as informative as being high.

For many applications, decision rules employing cutoff points are needed, and it is certainly possible empirically to establish useful cutoffs on one or more NEO-PI-3 scales (e.g., Costa & McCrae, 2005). Generally, however, the rules generated will be valid only for the special purpose for which they were intended and should not replace the dimensional interpretation of the scales.

Choice of Comparison Group

Raw scores on personality inventories may reflect social norms or stereotypes, but for personality assessment, they are usually meaningless—responses take on meaning only when they are compared to the responses of others. Published norms are intended to serve as the standard reference group, but the choice of appropriate norms requires some thought by the professional. Traditionally, separate norms have been used for men and women, so that a *T* score of 50 means average for a man or for a woman. For some applications, such as job selection, the user may want to compare the individual to people in general. For this reason, combined-sex normative information from which *T* scores can be calculated is provided in Appendices B and E. Separate Profile Forms based on combined-gender norms are available.

Normally it is appropriate to use adolescent norms to interpret the scores of individuals age 12 to 20 years. Similarly, it would be possible to plot the scores of adolescents using adult norms to see how they compare to adults in general. Many adolescents would score quite high on E5: Excitement-Seeking; this is meaningful because adolescents are typically higher than adults, in general, in seeking stimulation. A more detailed discussion of the choice of normative groups is given in McCrae, Martin, and Costa (2005).

Standard instructions suggest using Profile Forms for adult norms for interpreting the scores of an individual age 21 to 30 years. However, because these young adults tend to obtain scores that are intermediate between those obtained by adolescents and older adults, it may sometimes be appropriate to use the Adolescent Profile Forms instead.

Either choice is appropriate as long as the professional understands that when the Adult Profile Form is being used, the individual is being described in comparison with the average adult; when the Adolescent Profile Form is used, the individual is being described in comparison with the average adolescent. Appendices E.1 and E.2 report separate norms for adults age 21 to 30 years for those who wish to use the most specific age group, but there are no published Profile Forms using these norms.

Clinical Profiles

For several reasons, separate Profile Forms have not been developed for clinical groups. First, patients in psychotherapy generally show the same range of variation in personality traits as do other adults, differing only in the mean level of some traits (e.g., elevated *N* scores). Second, specific diagnoses may be associated with particular profiles, and these profiles can be seen most clearly when they are plotted against the normal adult profile. Finally, we believe that the use of nonclinical norms is a valuable reminder that individuals in psychotherapy are similar in many respects to normal volunteers.

The Five-Factor Model of Personality

The NEO Inventories were developed to operationalize the Five-Factor Model of personality, a representation of the structure of traits that was developed and elaborated over many years (Digman, 1990; John, Naumann, & Soto, 2008). The five factors represent the most basic dimensions underlying the traits identified in both natural languages and psychological questionnaires.

One major line of research—the lexical tradition (John, Angleitner, & Ostendorf, 1988)—began with an analysis of trait adjectives found in English and other natural languages. Words like *nervous*, *energetic*, *original*, *accommodating*, and *careful* evolved over the course of centuries to allow individuals to describe themselves and others. Thousands of such words are found in the dictionary, and trait theorists like Cattell (1946) and Norman (1963) proposed that this list of terms could be considered an exhaustive enumeration of personality traits. By factor analyzing ratings on all these adjectives, they argued, one should uncover the structure of personality traits themselves. Through a series of studies, this research led to the identification of the five factors (John et al., 2008).

Although derived from an analysis of lay terms, these factors were familiar to personality psychologists who had studied similar traits. Since 1985, research using the NEO Inventories has demonstrated that the same five factors can account for the major dimensions in personality questionnaires designed to measure Jungian functions, Murray's

needs, the traits of the Interpersonal Circumplex, and *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition (DSM-IV)*; American Psychiatric Association, 1994) personality disorders. It appears that these factors are indeed comprehensive (McCrae & Costa, 2008a).

Factors are defined by groups of intercorrelated traits. We refer to these more specific traits as *facets*, and to each cluster of facets as a *domain*. Summing the facet scales yields the domain score, which can be thought of as an approximation to the factor score. Factor scores are calculated directly by the NEO Software System.

By describing the individual's standing on each of the five factors, the NEO Inventories provide a comprehensive sketch that summarizes an individual's emotional, interpersonal, experiential, attitudinal, and motivational styles. NEO-PI-3 domain scales and factors measure personality at this level; facet scales offer a more fine-grained analysis by measuring specific traits within each of the five domains.

The Five Domains

The first step in interpreting a NEO-PI-3 profile is to examine the five domain scales to understand personality at the broadest level. This section describes each of the domains or factors and presents basic definitions as well as crucial distinctions.

Neuroticism (N)

The most pervasive domain of personality scales contrasts adjustment or emotional stability with maladjustment or Neuroticism (N). Although clinicians distinguish among many different kinds of emotional distress, from social phobia to agitated depression to borderline hostility, innumerable studies have shown that individuals prone to any one of these emotional states are also likely to experience others (Costa & McCrae, 1992a). The general tendency to experience negative affects such as fear, sadness, embarrassment, anger, guilt, and disgust is the core of the N domain. However, N includes more than susceptibility to psychological distress. Perhaps because disruptive emotions interfere with adaptation, men and women high in N are also prone to have irrational ideas, to be less able to control their impulses, and to cope more poorly with stress than others.

As the name suggests, patients traditionally diagnosed as suffering from neuroses generally score higher than others on measures of N (e.g., Eysenck & Eysenck, 1964). But the N scale of the NEO-PI-3, like all NEO-PI-3 scales, measures a dimension of general personality. High scorers may be at risk for some kinds of psychiatric problems, but

the N scale should not be viewed as a measure of psychopathology. It is possible to obtain a high score on the N scale without having any diagnosable psychiatric disorder. Conversely, not all psychiatric categories imply high levels of N. For example, an individual may have an antisocial personality disorder without having an elevated N score.

Individuals who score low on N are emotionally stable. They are usually calm, even-tempered, and relaxed, and they are able to face stressful situations without becoming upset or rattled.

Extraversion (E)

Extraverts are, of course, sociable, but sociability is only one of the traits that comprise the domain of Extraversion (E). In addition to liking people and preferring large groups and gatherings, extraverts are also assertive, active, and talkative. They like excitement and stimulation and tend to be cheerful in disposition. They are upbeat, energetic, and optimistic. Salespeople represent the prototypic extraverts in our culture, and the E domain scale is strongly correlated with interest in enterprising occupations (Costa, McCrae, & Holland, 1984; De Fruyt & Mervielde, 1997).

Although it is easy to convey the characteristics of the extravert, the introvert is less easy to portray. In some respects, introversion should be seen as the absence of Extraversion rather than what might be assumed to be its opposite. Thus, introverts are reserved rather than unfriendly, independent rather than followers, and even-paced rather than sluggish. Introverts may say they are shy when they mean they prefer to be alone: They do not necessarily suffer from social anxiety. Finally, although they are not given to the exuberant high spirits of extraverts, introverts are not unhappy or pessimistic. Curious as some of these distinctions may seem, they are strongly supported by research and form one of the most important conceptual advances of research on the Five-Factor Model (Costa & McCrae, 1980a; McCrae & Costa, 1987). Breaking the mental sets that link such pairs as happy/unhappy, friendly/hostile, and outgoing/shy allows important new insights into personality.

Users familiar with Jungian psychology should note that the conceptualization of E embodied in the NEO Inventories differs in many respects from Jung's (1923/1971) theory. In particular, introspection or reflection is not related to either pole of E, but is instead a characteristic of individuals who score high on Openness to Experience. Further discussion of these points is provided in McCrae and Costa (1989a).

Openness to Experience (O)

As a major dimension of personality, Openness to Experience (O) is much less well known than N or E. The elements of O—active imagination, aesthetic sensitivity, attentiveness to inner feelings, preference for variety, intellectual curiosity, and independence of judgment—have often played a role in theories and measures of personality, but their coherence into a single broad domain was not recognized for many years. The O scale of the NEO Inventories is perhaps the most widely researched measure of this broad domain (McCrae, 1996; McCrae & Sutin, 2009).

Open individuals are curious about both inner and outer worlds, and their lives are experientially richer than those of closed individuals. They are willing to entertain novel ideas and unconventional values, and they experience both positive and negative emotions more keenly than do closed individuals.

Alternative formulations of the Five-Factor Model often label this factor *Intellect*, and O scores are modestly associated with both education and measured intelligence. Openness is especially related to aspects of intelligence—such as divergent thinking—that contribute to creativity (McCrae, 1987). But Openness is by no means equivalent to intelligence. Some very intelligent people are closed to experience, and some very open people are quite limited in intellectual capacity. In a factor analytic sense, measures of cognitive ability form a sixth independent factor that we regard as being outside the domain of personality proper (McCrae, 1994b).

Men and women who score low on O tend to be conventional in behavior and conservative in outlook. They prefer the familiar to the novel, and their emotional responses are somewhat muted. Although an individual's level of openness may influence the form of psychological defense he or she uses (McCrae & Costa, 1997a), there is no evidence that closedness itself is a generalized defensive reaction. Instead, it seems likely that closed people simply have a narrower scope and a lower intensity of interests. Similarly, although they tend to be socially and politically conservative, closed people should not be viewed as authoritarians. Closedness does not imply hostile intolerance or authoritarian aggression. These qualities are more likely to be signs of extremely low Agreeableness.

A related distinction must be made at the open pole. Open individuals are unconventional, willing to question authority, and prepared to entertain new ethical, social, and political ideas. These tendencies, however, do not mean that they are unprincipled. An open person may apply his

or her evolving value system as conscientiously as a traditionalist does.

Openness may sound healthier or more mature to many psychologists, but the value of openness or closedness depends on the requirements of the situation, and both open and closed individuals perform useful functions in society.

Agreeableness (A)

Like Extraversion, Agreeableness (A) is primarily a dimension of interpersonal tendencies. The agreeable person is fundamentally altruistic. He or she is sympathetic to others and eager to help them, and believes that others will be equally helpful in return. By contrast, low scorers on A, disagreeable or antagonistic people, are egocentric, skeptical of others' intentions, and competitive rather than cooperative.

It is tempting to see the agreeable side of this domain as both socially preferable and psychologically healthier, and it is certainly the case that agreeable people are more popular than antagonistic individuals. However, the readiness to fight for one's own interests is often advantageous, and Agreeableness is not a virtue on the battlefield or in the courtroom. Skeptical and critical thinking contributes to accurate analysis in the sciences.

Just as neither pole of this dimension is intrinsically better than the other from society's point of view, neither pole is necessarily better in terms of the individual's mental health. Horney (1945) discussed two neurotic tendencies—moving toward people and moving against people—that resemble pathological forms of Agreeableness and antagonism. Low A is associated with narcissistic, antisocial, and paranoid personality disorders, whereas high A is associated with the dependent personality disorder (Costa & Widiger, 2002).

Conscientiousness (C)

A great deal of personality theory, particularly psychodynamic theory, concerns the control of impulses. During the course of development, most individuals learn how to manage their desires, and the inability to resist impulses and temptations is generally a sign of high N scores among adults. But self-control also can refer to a more active process of planning, organizing, and carrying out tasks; individual differences in this tendency are the basis of Conscientiousness (C; Costa & McCrae, 1998b; McCrae & Löckenhoff, 2010).

The conscientious individual is purposeful, strong-willed, and determined, and probably few people become great musicians or athletes without a reasonably high level of this trait. Digman and Takemoto-Chock (1981) referred

to this domain as *Will to Achieve*. On the positive side, high C is associated with academic and occupational achievement; on the negative side, it may lead to annoying fastidiousness, compulsive neatness, or workaholic behavior.

Conscientiousness is an aspect of what was once called *character*: high C scorers are scrupulous, punctual, and reliable. Low scorers are not necessarily lacking in moral principles, but they are less exacting in applying them, just as they are more lackadaisical in working toward their goals. There is some evidence that low scorers are more hedonistic and interested in sex than high scorers (McCrae, Costa, & Busch, 1986).

The Facet Scales

Each of the five domains of the NEO-PI-3 is represented by six more specific scales that measure facets of the domain. There are several advantages to the strategy of assessing a variety of facets. First, it ensures that the items used to measure the domain will cover as wide a range of relevant thoughts, feelings, and actions as possible. The N scale, for example, must include items measuring anger, depression, self-consciousness, impulsiveness, and vulnerability to stress, as well as anxiety. Domain scores are thus designed to reflect the broadest possible dimensions of personality.

Second, having several independent facet scales permits internal replication of findings. For example, *each* of the six facets of N is significantly related to negative affect and lower life satisfaction (Costa & McCrae, 1984), which gives considerable confidence to the claim that N is indeed related to psychological well-being. Similarly, the clinician who sees that a patient is high in anxiety, hostility, self-consciousness, and depression can be confident that he or she has pervasive psychological distress.

A third, and crucial, advantage to the multifaceted approach to the measurement of the five factors arises from the fact that meaningful individual differences can be seen *within* domains. Openness to fantasy, aesthetics, feelings, actions, ideas, and values covary to define the Openness factor, and individuals who score high on one facet are likely to score high on others. But this is only a statement of probability. Some individuals, for example, are open to new ideas but not values, or are open to feelings but not aesthetics. These individual differences within domains are stable over time and confirmed by observer ratings (McCrae & Costa, 1990, 1992), so they must be regarded as real facts of personality and not merely random scatter. Analyses of data from twins show that the specific variance associated with individual facet scales is

substantially heritable (Jang, McCrae, Angleitner, Riemann, & Livesley, 1998).

Examination of facet scales can provide a more fine-grained analysis of persons or groups. This can be particularly illuminating when the overall domain score is in the average range. For example, an individual whose average A score includes very low Altruism but very high Compliance will react quite differently from an individual with an equal A score but with a pattern of high Altruism and low Compliance.

Finally, the detailed information available from consideration of facet scores can be useful in interpreting constructs and formulating theories. Extraversion is known to be related to psychological well-being (Costa & McCrae, 1984), but a closer look shows that two of the facets, Warmth and Positive Emotions, are chiefly responsible for this association; Excitement-Seeking is not related to well-being. Such findings have important implications for a theory of well-being.

Neuroticism Facets

N1: Anxiety

Anxious individuals are apprehensive, fearful, prone to worry, nervous, tense, and jittery. The scale does not measure specific fears or phobias, but high scorers are more likely to have such fears, as well as free-floating anxiety. Low scorers are calm and relaxed. They do not dwell on things that might go wrong.

N2: Angry Hostility

Angry hostility represents the tendency to experience anger and related states such as frustration and bitterness. This scale measures the individual's readiness to *experience* anger; whether the anger is *expressed* depends on the individual's level of Agreeableness. Note, however, that disagreeable people often score high on this scale. Low scorers are easy-going and slow to anger.

N3: Depression

This scale measures individual differences in the tendency to experience depressive affect. High scorers are prone to feelings of guilt, sadness, hopelessness, and loneliness. They are easily discouraged and often dejected. Low scorers rarely experience such emotions, but they are not necessarily cheerful and lighthearted—characteristics associated instead with Extraversion.

N4: Self-Consciousness

The emotions of shame and embarrassment form the core of this facet of N. Self-conscious individuals are uncomfortable around others, sensitive to ridicule, and prone to feelings of inferiority. Self-consciousness is akin to shyness and social anxiety—to Fenigstein, Scheier, and

Buss' (1975) public (but not private) self-consciousness. Low scorers do not necessarily have poise or good social skills, they are simply less disturbed by awkward social situations.

N5: Impulsiveness

In the NEO-PI-3, impulsiveness refers to the inability to control cravings and urges. Desires (e.g., for food, cigarettes, possessions) are perceived as being so strong that the individual cannot resist them, although he or she may later regret the behavior. Low scorers find it easier to resist such temptations and have a high tolerance for frustration. The term *impulsive* is used by many theorists to refer to many different and sometimes unrelated traits (see Whiteside & Lynam, 2001). The NEO-PI-3 N5: Impulsiveness facet should not be confused with spontaneity, risk-taking, or rapid decision time.

N6: Vulnerability

The final facet of N is vulnerability to stress. Individuals who score high on this scale feel unable to cope with stress, becoming dependent, hopeless, or panicked when facing emergency situations. Low scorers perceive themselves as capable of handling themselves in difficult situations.

Extraversion Facets

E1: Warmth

Warmth is the facet of Extraversion most relevant to issues of interpersonal intimacy. Warm people are affectionate and friendly. They genuinely like people and easily form close attachments to others. Low scorers are neither hostile nor necessarily lacking in compassion, but they are more formal, reserved, and distant in manner than high scorers. Warmth is the facet of E that is closest to Agreeableness in terms of interpersonal space, but it is distinguished by a cordiality and heartiness that is not part of A.

E2: Gregariousness

A second aspect of E is gregariousness: the preference for other people's company. Gregarious people enjoy the company of others—"the more the merrier." Low scorers on this scale tend to be loners who do not seek—or who even actively avoid—social stimulation.

E3: Assertiveness

High scorers on this facet are dominant, forceful, and socially ascendant. They speak without hesitation and often become group leaders. Low scorers prefer to keep in the background and let others do the talking.

E4: Activity

A high Activity score is seen in rapid tempo and vigorous movement, a sense of energy, and a need to keep busy.

Active people lead fast-paced lives. Low scorers are more leisurely and relaxed in tempo, though they are not necessarily sluggish or lazy.

E5: Excitement-Seeking

High scorers on this scale crave excitement and stimulation. They like bright colors and noisy environments. Excitement-Seeking is akin to some aspects of sensation seeking (Zuckerman, 1979). Low scorers feel little need for thrills and prefer a lifestyle that high scorers might find boring.

E6: Positive Emotions

The last facet of E assesses the tendency to experience positive emotions such as joy, happiness, love, and excitement. High scorers on the Positive Emotions scale laugh easily and often. They are cheerful and optimistic. Low scorers are not necessarily unhappy; they are merely less exuberant and high-spirited. Research (e.g., Schimmack, Oishi, Furr, & Funder, 2004; Steel, Schmidt, & Shultz, 2008) has shown that happiness and life satisfaction are related to both N and E, and that Positive Emotions is the facet of E most relevant to the prediction of happiness.

Openness Facets

By convention, facets of O are designated by the aspect or area of experience to which the individual is open. Thus, a high scorer on the Fantasy scale enjoys rich, varied, and novel experiences in his or her fantasy life; a high scorer on the Ideas facet enjoys rich, varied, and novel experiences in his or her intellectual life. In publications, the implicit "open to..." is usually expressed. Thus, McCrae and Costa (1989b) wrote that "the MBTI [Myers-Briggs Type Indicator] TF [Thinking-Feeling] scale... was directly related to *Openness* [emphasis added] to Feelings" (p. 32).

O1: Fantasy

Individuals who are open to fantasy have a vivid imagination and an active fantasy life. They daydream not simply as an escape, but as a way of creating an interesting inner world for themselves. They elaborate and develop their fantasies and believe that imagination contributes to a rich and creative life. Low scorers are more prosaic and prefer to keep their minds on the task at hand.

O2: Aesthetics

High scorers on this scale have a deep appreciation for art and beauty. They are moved by poetry, absorbed in music, intrigued by art, and they often experience a chill in response to sudden beauty (McCrae, 2007). They need not have artistic talent, nor even necessarily what most people would consider good taste, but for many of them, their interest in the arts will lead them to develop a wider knowledge and appreciation than that of the average individual.

Low scorers are relatively insensitive to and uninterested in art and beauty.

O3: Feelings

Openness to feelings implies receptivity to one's own inner feelings and emotions and the evaluation of emotion as an important part of life. High scorers experience deeper and more differentiated emotional states and feel both happiness and unhappiness more keenly than others do. Low scorers have somewhat blunted affects and do not believe that feeling states are of much importance. They may be characterized by alexithymia.

O4: Actions

Openness is seen behaviorally in the willingness to try different activities, go new places, or eat unusual foods. High scorers on this scale prefer novelty and variety to familiarity and routine. Over time, they may engage in a series of different hobbies. Low scorers find change difficult and prefer to stick with the tried-and-true.

O5: Ideas

Intellectual curiosity is an aspect of Openness that has long been recognized (Fiske, 1949). This trait is seen not only in an active pursuit of intellectual interests for their own sake, but also in open-mindedness and a willingness to consider new, perhaps unconventional, ideas. High scorers enjoy both philosophical arguments and brain teasers. Openness to ideas does not necessarily imply high intelligence, though it can contribute to the development of intellectual potential. Low scorers on the scale have limited curiosity and, if highly intelligent, narrowly focus their resources on limited topics.

O6: Values

Openness to Values assesses the readiness to reexamine social, political, and religious values. Closed individuals tend to accept authority and to honor tradition and, as a consequence, are generally conservative, regardless of political party affiliation. Openness to Values may be considered the opposite of dogmatism (Rokeach, 1960).

Agreeableness Facets

A1: Trust

The first facet of Agreeableness is Trust. High scorers are disposed to believe that others are honest and well-intentioned. Low scorers on this scale tend to be cynical and skeptical and to assume that others may be dishonest or dangerous.

A2: Straightforwardness

Individuals with high scores on this scale are frank, sincere, and ingenuous. Low scorers on this scale are more willing to manipulate others through flattery, craftiness, or deception. They view these tactics as necessary social

skills and may regard more straightforward people as naïve. When interpreting this scale (as well as other A and C scales), the professional must remember that scores reflect standing *relative to other individuals*. A low scorer on this scale is more likely to stretch the truth or to be guarded in expressing his or her true feelings, but this should not be interpreted to mean that he or she is a dishonest or manipulative person. In particular, this scale should not be regarded as a lie scale, either for assessing the validity of the test itself or for making predictions about honesty in employment or other settings.

A3: Altruism

High scorers on the Altruism scale have an active concern for others' welfare as shown in generosity, consideration of others, and a willingness to assist others in need of help. Low scorers on this scale are somewhat more self-centered and are reluctant to get involved in the problems of others.

A4: Compliance

This facet of A concerns characteristic reactions to interpersonal conflict. The high scorer tends to defer to others, to inhibit aggression, and to forgive and forget. Compliant people are meek and mild. The low scorer is aggressive, prefers to compete rather than to cooperate, and has no reluctance to express anger when necessary.

A5: Modesty

High scorers on this scale are humble and self-effacing, though they are not necessarily lacking in self-confidence or self-esteem. Low scorers believe they are superior people and may be considered conceited or arrogant by others. A pathological lack of modesty is part of the clinical conception of narcissism.

A6: Tender-Mindedness

This facet scale measures attitudes of sympathy and concern for others. High scorers are moved by others' needs and emphasize the human side of social policies. Low scorers are more hardheaded and less moved by appeals to pity. They would consider themselves realists who make rational decisions based on cold logic.

Conscientiousness Facets

C1: Competence

Competence refers to the sense that one is capable, sensible, prudent, and effective. High scorers on this scale feel well-prepared to deal with life. Low scorers have a lower opinion of their abilities and admit that they are often unprepared and inept. Of all the C facet scales, Competence is most highly associated with self-esteem and internal locus of control (Costa, McCrae, et al., 1991). Not surprisingly, it is negatively related to N6: Vulnerability.

C2: Order

High scorers on this scale are neat, tidy, and well-organized. They keep things in their proper places. Low scorers are unable to get organized and describe themselves as unmethodical. Carried to an extreme, high Order may contribute to an obsessive-compulsive personality disorder.

C3: Dutifulness

In one sense, *conscientious* means "governed by conscience," and that aspect of C is assessed as Dutifulness. High scorers on this scale adhere strictly to their ethical principles and scrupulously fulfill their moral obligations as they understand them. Low scorers are more casual about such matters and may be somewhat undependable or unreliable.

C4: Achievement Striving

Individuals who score high on this facet have high aspiration levels and work hard to achieve their goals. They are diligent and purposeful and have a sense of direction in life. Very high scorers, however, may invest too much in their careers and become workaholics. Low scorers are lackadaisical and perhaps even lazy. They are not driven to succeed. They lack ambition and may seem aimless, but they are often perfectly content with their low levels of achievement.

C5: Self-Discipline

By this term, we refer to an individual's ability to begin tasks and carry them through to completion, despite boredom or other distractions. High scorers can motivate themselves to get the job done. Low scorers procrastinate in beginning chores and are easily discouraged and eager to quit. Low self-discipline is easily confused with impulsiveness—both are evidence of poor self-control—but empirically, they are distinct. People who have high levels of impulsiveness cannot resist doing what they do not want themselves to do; people who have low levels of self-discipline cannot force themselves to do what they want themselves to do. The former requires an emotional stability; the latter, a degree of motivation that they do not possess.

C6: Deliberation

The final facet of C is deliberation—the tendency to think carefully before acting. High scorers on this facet are cautious and deliberate. Low scorers are hasty and often speak or act without considering the consequences. At best, low scorers are spontaneous and able to make snap decisions when necessary.

Interpreting Profiles

An individual's set of scores can be plotted on a Profile Form to see the overall configuration of his or her

personality relative to the appropriate normative group. The most distinctive and salient domains and facets can then be identified. Familiarity with the correlates of NEO-PI-3 scales allows the interpreter to make predictions about important aspects of the individual's life, such as coping styles, vocational interests, and life satisfaction.

Profile interpretation can be as simple as noting a few distinctive traits or as complex as the knowledge, skill, and interest of the interpreter permit. In clinical use, the NEO-PI-3 Profile Form should be considered in the context of the client's history, presenting problems, and other psychological test data, such as spouse ratings on Form R of the NEO-PI-3. In some cases, it may be appropriate to discuss the profile with the client, a process that can help both client and therapist understand the ways these personality traits are manifested in the client's life. As with all psychological test results, NEO-PI-3 profile interpretations must be considered tentative. Neither self-reports nor observer ratings (nor, for that matter, clinical judgments) are infallible. If test results appear inaccurate to the therapist or client, further exploration is indicated.

Making sense of 30 unrelated scales would be extremely difficult, and thus we have recommended that interpreters examine the facets domain by domain. However, there are also other ways of thinking about facet scales that may be useful, especially for an experienced interpreter. One way is to consider the relations of facet scales across domains—an approach that is justified by the secondary factor loadings of several scales (see Table 7 in chapter 8 of this Professional Manual). For example, antagonistic people, in addition to scoring low on the six A facets, also tend to score high on N2: Angry Hostility and to score low on E1: Warmth. In gauging the depth and pervasiveness of the individual's antagonism, consideration of all these scales is useful. Similarly, the meaning of a high O1: Fantasy score may depend on other scores. In an individual who is otherwise closed to experience, fantasy may be used as an escape from stress, or as a way of avoiding unpleasant tasks. High N scores would support the former interpretation; low C scores would suggest the latter.

Combinations of Traits

It also may be useful to consider pairs of domain or factor scores in terms of two-dimensional planes, which correspond, in many cases, to particular areas of life. Two of these combinations have been extensively researched: the affective plane, defined by N and E, which represents the individual's basic emotional styles (Costa & McCrae, 1980a; Watson & Tellegen, 1985); and the interpersonal plane, or circumplex, defined by E and A (McCrae & Costa, 1989d; Costa & McCrae, in press). E and O together are important both for vocational interests (Costa

et al., 1984) and for the selection of optimal forms of therapy (T. Miller, 1991).

The 10 NEO Style Graphs describe all combinations of pairs of factors: Styles of Well-Being (N and E), Defense (N and O), Anger Control (N and A), Impulse Control (N and C), Interests (E and O), Interactions (E and A), Activity (E and C), Attitudes (O and A), Learning (O and C), and Character (A and C). Interpretation of these 10 styles is included in the NEO-PI-3 Software System Interpretive Report and in the NEO Style Graph Booklet.

More complex combinations of traits also can be considered. In particular, *DSM-IV* personality disorders can be conceptualized in terms of discrete combinations of traits. For example, the paranoid personality disorder is associated with the traits of N2: Angry Hostility, low A1: Trust, low A2: Straightforwardness, and low A4: Compliance. Individuals who show this pattern of traits may meet the diagnostic criteria for paranoid personality disorder. The *Clinical Hypotheses* section of the computer-generated interpretive report compares the respondent's profile to established patterns for each of the personality disorders and indicates which diagnoses are likely or unlikely. NEO-PI-3 profiles can also be compared to many other empirically-based profiles (such as those for patients with Alzheimer's disease [Siegler et al., 1991] or those for prototypic psychopaths [Miller, Lynam, Widiger, & Leukefeld, 2001]) to assess similarity of personality patterns.

Combining Self-Reports and Observer Ratings

When both Form S and Form R data are available for an individual, both the complexities and possibilities of profile interpretation are multiplied. The profiles can be interpreted separately to understand personality as it is seen by the individual and as it is seen by those around him or her. Alternatively, two (or more) profiles can be plotted on the same form (using *T* scores based on relevant norms), and the interpretations can focus on points of agreement and disagreement.

The clinical interpretation of joint profiles—particularly their discrepancies—has been discussed by Mutén (1991) and Singer (2005), and an extended case study is provided by Costa and Piedmont (2003). Formal statistical analyses indicating the degree of agreement and pointing to specific scores that require further information gathering have been developed (McCrae, 1994a) and form the basis of the interpretation of joint profiles in the NEO Software System Combined Report.

An Example of Profile Interpretation: Case A

To illustrate profile interpretation, Figure 3 presents the NEO PI-R profile of a 32-year-old married woman

enrolled in a behavioral medicine clinic. (Note that the interpretation would be identical if she had received the same *T* scores on the NEO-PI-3.) Case A came to the clinic complaining of pain in her upper back and neck, financial worries, and stress. She was a professional in a high-technology industry that had recently suffered losses, and she had reason to believe that she might be laid off from work. She had some previous experience with psychotherapy.

At the left-hand side of the Profile Form, the five domain scores are plotted. From these, one can see that this individual views herself as near average on N and E, and high on O, A, and C. Moving to the right, the facet scale scores for each domain are plotted. Note that there is considerable scatter within the N domain. This woman considers herself to be very high in N1: Anxiety and high in N3: Depression and N6: Vulnerability, but very low in N5: Impulsiveness—perhaps as a result of her high level of Conscientiousness. She is average to high in most facets of E, but clearly low in E6: Positive Emotions. She is average to high in all facets of O, A, and C except for O1: Fantasy and C4: Achievement Striving, on both of which she scores very high.

This is the profile of an individual with a good prognosis: moderate levels of N and high levels of C are associated with good treatment outcomes (Miller, 1991). Therapy included promoting self-nurturing behaviors (to increase positive emotions) and learning relaxation skills (to reduce anxiety). In a series of seven treatment sessions, she made substantial progress in reducing her back pain and in making constructive plans for her future that would allow her to satisfy her needs for achievement without undue stress.

The NEO-PI-3 Interpretive Report

Both the NEO Software System and the NEO Professional Report Service generate reports that can be used to illustrate profile interpretation. Naturally, these reports deal only with NEO Inventory data; it remains the responsibility of the professional to integrate this information with other relevant facts about the client.

The report first gives identifying information about the client and produces the client's NEO-PI-3 profile. A listing of the raw scores, *T* scores, and ranges for the factors and facet scales is then presented. Norms for these scores are based on the age and sex of the individual and on the form of the test that was administered. Either within-sex or combined-sex norms can be chosen.

The program then evaluates the validity of the responses, taking into account missing item responses, acquiescence, nay-saying, random responding, and the answers to the validity check items. If 41 or more item

Females Form S (Self Report)

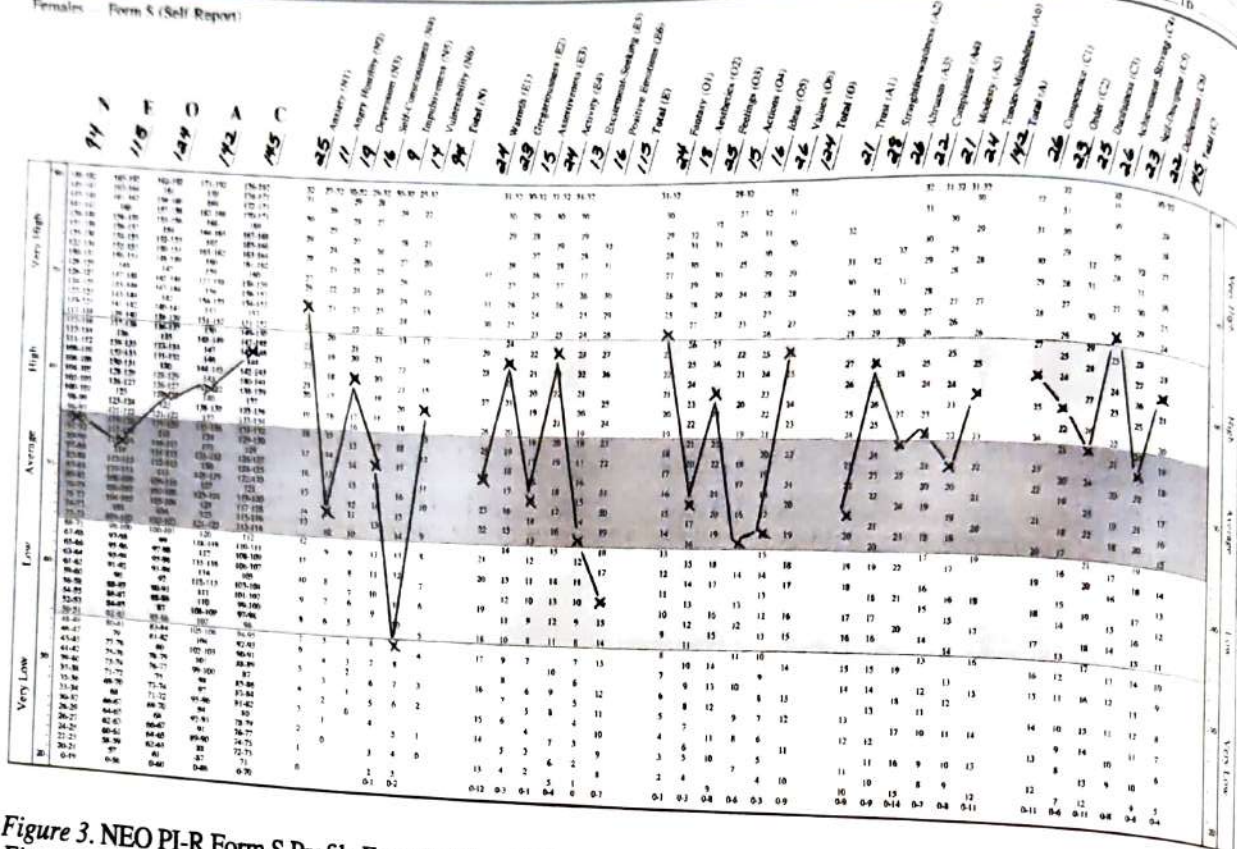


Figure 3. NEO PI-R Form S Profile for Case A. From the *Revised NEO Personality Inventory (NEO PI-R) and NEO Five-Factor Inventory (NEO-FFI) Professional Manual* (p. 20), by P. T. Costa, Jr. & R. R. McCrae, 1992, Odessa, FL: PAR. Copyright 1992 by PAR.

responses are missing, no report is generated. Otherwise, the report is produced along with various cautionary statements derived from the outcome of the validity evaluation. After reading the *Validity Indices* section of the report, the professional must decide whether to obtain more information from the respondent, discard the test, or retain the interpretation with the appropriate cautions.

The *Basis of Interpretation* section identifies the source of the NEO-PI-3 data and the normative group selected for comparison. It also provides information on the five factor scores that are used as the basis of the global interpretation of personality.

The interpretive material is divided into three main sections. The first, *Global Description of Personality: The Five Factors*, describes the respondent on each factor, beginning with the most distinctive factor score (the furthest from average), using decision logic that classifies scores into five levels of elevation (from Very Low to Very High). The second section, *Detailed Interpretation: Facets of N, E, O, A, and C*, contains narratives based on facet scores. The last major section of the basic report, *Personality*

Correlates: Some Possible Implications, draws on empirical research using the NEO PI-R to broaden the interpretation. Included are statements on coping and defenses, somatic complaints, psychological well-being, cognitive processes, interpersonal characteristics, and needs and motives. Finally, the report can include relevant sections of the NEO Problems in Living Checklist, based on the client's NEO scores. This section lists personality-related difficulties the client may be facing and can serve as a guide to a focused interview.

There are several optional sections of the report. A Client Report, *Your NEO-PI-3 Summary*, is available for providing detailed feedback based on self-reports. The NEO Style Graph Booklet can also be generated. For clients in psychotherapy, an optional *Clinical Hypotheses* section addresses possible Axis II diagnoses (for adults) and lists some treatment implications of the profile. If data are available from two ratings of the same target, a Combined Report can be created, based on the adjusted mean score (see McCrae, Stone, Fagan, & Costa, 1998 for details on the statistical adjustment).

Interpretive reports are generated by applying a set of decision rules to the profile of scores: If the scores exceed certain cutoff points, statements that summarize the meaning and possible implications of the scores are included in the report. The statements are based on scale content and on research findings; the cutoff points were chosen based on the judgment and experience of the authors. These interpretive reports should be viewed as a set of probabilistic inferences that an interpreter who is well-versed in the literature on the NEO Inventories might make given the observed scores. To document the research basis of the statements about personality correlates and consequences, references are given in brackets in the case studies that follow. Note that these references do not appear in the computer generated reports.

Interpretive reports such as these, which compile a portrait through the application of a series of decision rules, sometimes appear to be incongruous. For example, in the global description of Case B, it is stated that introverts like him "avoid large, noisy parties," which is, in general, true. However, Case B happens to be average on E2: Gregariousness, so the detailed description states that he "sometimes enjoys large and noisy crowds or parties." Apparent inconsistencies in interpretive reports should not be seen as errors but as issues that require additional clarification and further individualized interpretation. *If there are discrepancies between the domain and facet interpretations, the more specific facet interpretations should normally be given more weight.*

Case Studies

Four additional cases studies are presented; all provided appropriate informed consent. The first (Case B) illustrates a simple self-report profile for an individual whose most notable characteristics are a Very High level of N1:

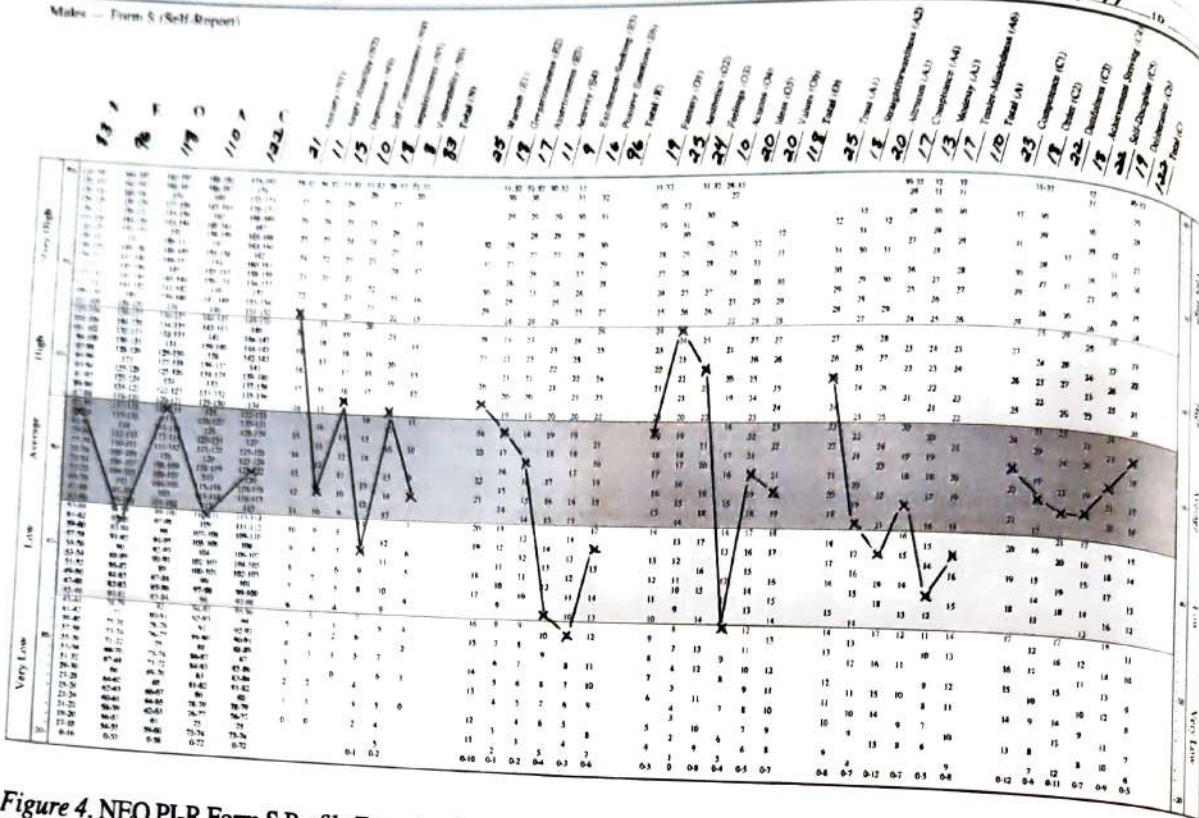
Anxiety and Very Low levels of E5: Excitement-Seeking and O4: Openness to Actions. Case B is intended to illustrate the basic interpretive report as well as one of the 10 NEO Style Graphs. The second example (Case C) combines information from an autobiography, ratings from two peers, and a narrative based on one of the peers' ratings. In addition to the basic report, the *Clinical Hypotheses* section is presented to suggest possible personality disorder diagnoses that might be considered if Case C were a patient in psychotherapy. Both of these cases are, in fact, normal volunteers (McCrae, 1994a). The third case (Case D) is a patient from a clinician in private practice; this case study illustrates the joint examination of self-reports and spouse ratings and the use of NEO PI-R data in clinical case management. For Case D, we also present *Your NEO PI-R Summary*, a Client Report that is more detailed than Your NEO Summary. It is based on the self-report only. Finally, Case E is based on a clinician's NEO-PI-3 rating of a patient in therapy and illustrates the use of the new NEO Problems in Living Checklist described in chapter 6 of this Professional Manual.

When comparing the profiles presented on the NEO-PI-3 Profile Forms with the text of the interpretive reports, users must recall that the text is based on factor *T* scores rather than on domain *T* scores. This difference accounts for apparent inconsistencies between the plotted profiles and the interpretive text. For example, Case B's domain score for Openness ($T = 55$) is in the Average range, but his factor score ($T = 56$) is in the High range.

Case B

Figure 4 shows the NEO PI-R Form S profile for a 79-year-old man. After studying literature in graduate school, he became a teacher and later a principal. He retired at age 75.

Males - Form S (Self-Report)



His profile is interpreted as follows:

—Validity Indices—

Validity indices (i.e., A and C questions) are within normal limits.

Because the NEO PI-R™ Scale Raw Score entry option was used, no checks for missing items, acquiescence, nay-saying, or random responding could be made.

—Basis of Interpretation—

This report compares the respondent to other adult men. It is based on self-reports of the respondent.

At the broadest level, personality can be described in terms of five basic dimensions or factors. NEO PI-R domain scores provide good estimates of these five factors by summing the six facets in each domain. Domain scores can be calculated easily by hand and are therefore used on the (hand-scored) Profile Form. More precise estimates of standing on the five factors, however, are provided by factor scores, which are a weighted combination of scores on all 30 facets (see Table 4 in the Manual). Factor scores are best calculated by computer.

Because factor scores have somewhat higher convergent and discriminant validity, they are used as the basis of this report. In general, domain *T* scores and factor *T* scores are very similar; occasionally, however, they differ. In these cases, the factor *T* score, which incorporates information from all 30 facets, is usually a more accurate description of the individual.

Factor scores are used to describe the individual at a global level, based on a composite of facet scale scores. To the extent that there is wide scatter among facet scores within a domain, interpretation of that domain and factor becomes more complex. Interpretive statements at the factor level may occasionally conflict with interpretive statements at the facet level. In these cases, particular attention should be focused on the facet scales and their interpretations.

—Global Description of Personality: The Five Factors—

The most distinctive feature of this individual's personality is his standing on the factor of Extraversion. Such people are somewhat introverted, preferring to do many things alone or with a small group of people. They avoid large, noisy parties and tend to be quiet and reserved in social interactions. Those who know such people would probably describe them as retiring and serious [McCrae & Costa, 1987]. The fact that these individuals are introverted does not necessarily mean that they lack social skills—many introverts function very well in social situations, although they might prefer to avoid them. Note also that introversion does not imply introspection; these individuals are likely to be thoughtful and reflective only if they are also high in Openness.

This person is high in Openness. High scorers like him are interested in experience for its own sake. They enjoy novelty and variety. They are sensitive to their own feelings and have a greater than average ability to recognize the emotions of others. They have a high appreciation of beauty in art and nature. They are willing to consider new ideas and values, and may be somewhat unconventional in their own views. Peers rate such people as original and curious.

Next, consider the individual's level of Neuroticism. Individuals scoring in this range are average in terms of their emotional stability. They experience a normal amount of psychological distress and have a typical balance of satisfactions and dissatisfactions with life. They are neither high nor low in self-esteem. Their ability to deal with stress is as good as the average person's.

This person is average in Agreeableness. People who score in this range are about as good-natured as the average person. They can be sympathetic, but can also be firm. They are trusting but not gullible, and ready to compete as well as to cooperate with others.

Finally, the individual scores in the average range in Conscientiousness. Men who score in this range have a normal level of need for achievement. They are able to set work aside in pursuit of pleasure or recreation. They are moderately well organized and fairly reliable, and have an average amount of self-discipline.

—Detailed Interpretation: Facets of N, E, O, A, and C—

Each of the five factors encompasses a number of more specific traits, or facets. The NEO PI-R measures six facets in each of the five factors. An examination of the facet scores provides a more detailed picture of the distinctive way that these factors are seen in this person.

Neuroticism

This individual is anxious, generally apprehensive, and prone to worry. He sometimes feels frustrated, irritable, and angry at others and he is prone to feeling sad, lonely, and dejected. Embarrassment or shyness when dealing with people, especially strangers, is not a problem for him. He reports being poor at controlling his impulses and desires, but he is able to handle stress as well as most people.

Extraversion

This person is very warm and affectionate toward others and he sometimes enjoys large and noisy crowds or parties. He is as assertive as most men when the circumstances require. The individual has a low level of energy and prefers a slow and steady pace. Excitement, stimulation, and thrills have little appeal to him and he is less prone to experience feelings of joy and happiness than most men.

Openness

In experiential style, this individual is generally open. He has an average imagination and only occasionally daydreams or fantasizes. He is particularly responsive to beauty as found in music, art, poetry, or nature, and his feelings and emotional reactions are varied and important to him. He seldom enjoys new and different activities and has a low need for variety in his life. He has only a moderate level of intellectual curiosity and he is generally middle-of-the-road in his social, political, and moral beliefs.

Agreeableness

This person easily trusts others and usually assumes the best about anyone he meets. He is generally frank and sincere, but he tends to put his own needs and interests before others'. This individual holds his own in conflicts with others, but he is also willing to forgive and forget. He is quite proud of himself and his accomplishments, and happy to take credit for them. Compared to other people, he is hard-headed and tough-minded, and his social and political attitudes reflect his pragmatic realism.

Conscientiousness

This individual is reasonably efficient and generally sensible and rational in making decisions. He is moderately neat, punctual, and well organized, and he is reasonably dependable and reliable in meeting his obligations. He has a moderately high need for achievement, but he can also set work aside for recreation. He is average in self-discipline and generally finishes the tasks he starts. He is reasonably cautious, and generally thinks things through before acting.

—Personality Correlates: Some Possible Implications—

Research has shown that the scales of the NEO PI-R™ are related to a wide variety of psychosocial variables. These correlates suggest possible implications of the personality profile, because individuals who score high on a trait are also likely to score high on measures of the trait's correlates.

The following information is intended to give a sense of how this individual might function in a number of areas. It is not, however, a substitute for direct measurement. If, for example, there is a primary interest in medical complaints, an inventory of medical complaints should be administered in addition to the NEO PI-R.

Coping and Defenses

In coping with the stresses of everyday life, this individual is not very likely to react with ineffective responses, such as hostile reactions toward others, self-blame, or escapist fantasies [McCrae & Costa, 1986b]. He is more likely than most adults to use humor and less likely to use faith in responding to threats, losses, and challenges. In addition, he is somewhat less likely to use positive thinking and direct action in dealing with problems.

Somatic Complaints

This person likely responds in a normal fashion to physical problems and illness. He is prone neither to exaggerate nor to minimize physical symptoms and is fairly objective in assessing the seriousness of any medical problems that he might have [Costa & McCrae, 1980b, 1987a].

Psychological Well-being

Although his mood and satisfaction with various aspects of his life will vary with the circumstances, in the long run this individual is likely to experience the normal course of positive and negative feelings and be generally content with life. Because he is open to experience, his moods may be more intense and varied than those of the average man [Costa & McCrae, 1984; McCrae & Costa, 1991a].

Cognitive Processes

This individual is likely to be more complex and differentiated in his thoughts, values, and moral judgments than others of his level of intelligence and education [Lonky, Kaus, & Roodin, 1984]. He would also probably score higher on measures of ego development [McCrae & Costa, 1980]. Because he is open to experience, this individual is likely to perform better than average on tests of divergent thinking ability; that is, he can generate fluent, flexible, and original solutions to many problems [McCrae, 1987]. He may be considered creative in his work or hobbies.

Interpersonal Characteristics

Many theories propose a circular arrangement of interpersonal traits around the axes of Love and Status [Leary, 1957; Wiggins, 1979]. Within such systems, this person would likely be described as modest, submissive, cold, unfeeling, and especially aloof and reserved. His traits are associated with low standing on the interpersonal dimensions of Love and Status [McCrae & Costa, 1989c; Costa & McCrae, in press].

Needs and Motives

Research in personality has identified a widely used list of psychological needs. Individuals differ in the degree to which these needs characterize their motivational structure. The respondent is likely to show high levels of the following needs: affiliation, harm avoidance (avoiding danger), nurturance, and sentience (enjoyment of sensuous and aesthetic experiences). The respondent is likely to show low levels of the following needs: abasement, change, and play [Costa & McCrae, 1988a].

—Stability of Profile—

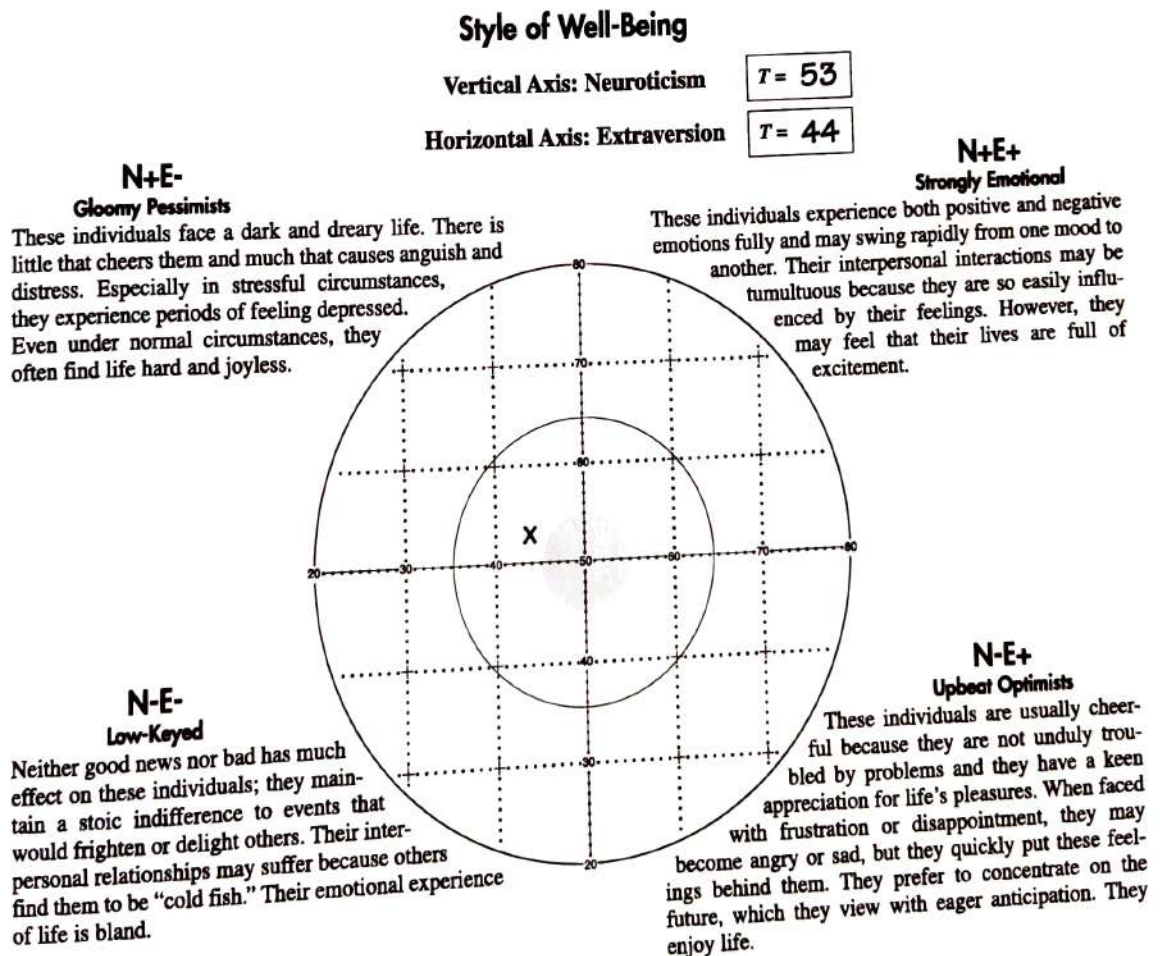
Research suggests that the personality profile of this respondent reflects not only current functioning, but will also be descriptive through the remainder of the respondent's life. Barring major changes in health

or therapeutic interventions, it is likely to continue to remain an accurate characterization [McCrae & Costa, 2003].

—Personality Style Graphs—

Broad personality factors are pervasive influences on thoughts, feelings, and actions, and combinations of factors provide insight into major aspects of people's lives, defining what can be called *personality styles*. For example, for many years psychologists have known that interpersonal interactions can be conceptualized in terms of a circular ordering or circumplex, defined by the two axes of Dominance and Love, or by the alternative axes of Extraversion and Agreeableness. These two factors define a *Style of Interactions*.

The nine other pairs of factors also define styles, and all ten are represented in NEO Style Graphs. An "X" is placed on each graph to indicate where the respondent falls; the description of that quadrant applies to the respondent. Descriptions are likely to be most accurate if (1) the "X" is far from the center; (2) the "X" is near the diagonal passing through the center of the quadrant; and (3) all the facets in each domain show similar levels. If the "X" is placed in the central circle, then none of the descriptions is especially relevant. If the "X" is located near the horizontal or vertical axis, then both quadrants on that side of the circle may be descriptive. If there is marked scatter among the facets in a domain, then interpretation should focus on these facets rather than the domain and its combinations in Style Graphs.



Case C

Case C is a 73-year-old woman who worked until age 60 while raising a family and taking care of her disabled father-in-law. Since retirement, she has been active as the leader of several social and civic groups, does volunteer work, gardens, does housework, and swims.

In describing her life, she frequently points to conflicts with others. As a young girl, she felt unappreciated at home. She "got a job as far away from home as possible and decided to fight for everything" she wanted out of life. She did not get along with her first employer, but because jobs were scarce during the Depression, she "fought to stay." Of her early married life she says, "We had plenty of battles... On occasion I did throw things at my

husband." And although she likes people and has many friends, she notes that "one doesn't like everybody he meets... I also distrust people who are sweet and insincere. Sweet and insincere are synonymous to me."

Figure 5 shows profiles from two peer raters of Case C plotted on a Form R Profile Form.

Note that although there are some differences, the two ratings are generally in agreement at both the domain level and the facet level. The following interpretive report is based on data from Rater 1 (the solid line). Although Case C is not a patient in psychotherapy, for illustrative purposes, this report includes the optional *Clinical Hypotheses* section that discusses personality disorders for which—if she were in psychotherapy—she might meet criteria.

NEO PI-R

Revised NEO Personality Inventory (NEO PI-R)

Paul T. Costa, Jr., Ph.D. and Robert R. McCrae, Ph.D.

Name Case C

Date _____ Age 73 ID _____

Females — Form R (Ratings)

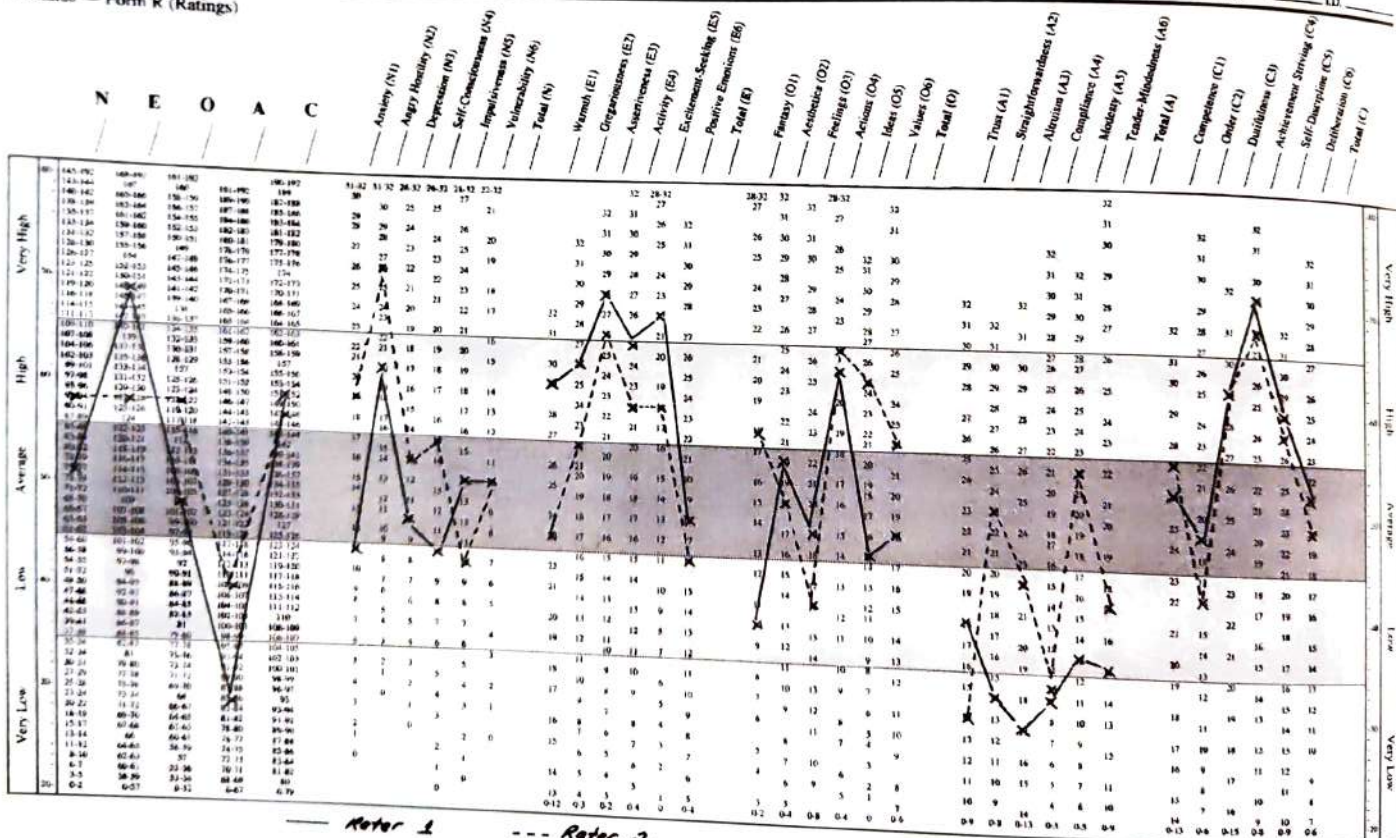


Figure 5. NEO PI-R Form R Profile Form showing ratings from two peers for Case C. From the *Revised NEO Personality Inventory (NEO PI-R) and NEO Five-Factor Inventory (NEO-FFI) Professional Manual* (p. 24), by P. T. Costa, Jr. & R. R. McCrae, 1992, Odessa, FL: PAR. Copyright 1992 by PAR.

—Basis of Interpretation—

This report compares this person to other adult women. It is based on observer ratings of the individual.

—Global Description of Personality: The Five Factors—

The most distinctive feature of this individual's personality is her standing on the factor of Agreeableness. People who score in this range are antagonistic and tend to be brusque or even rude in dealing with others. They are generally suspicious of other people and skeptical of others' ideas and opinions. They can be callous in their feelings. Their attitudes are tough-minded in most situations. They prefer competition to cooperation, and express hostile feelings directly with little hesitation. People might describe them as relatively stubborn, critical, manipulative, or selfish. (Although antagonistic people are generally not well-liked by others, they are often respected for their critical independence, and their emotional toughness and competitiveness can be assets in many social and business roles.)

This person is described as being high in Conscientiousness. Women who score in this range work toward their goals in a deliberate manner. They have a relatively high need for achievement. They are well organized and reliable and carry through on their commitments. They have good self-discipline and take their obligations seriously. Raters describe such people as careful and hardworking.

Next, consider the individual's level of Extraversion. Such people enjoy the company of others and the stimulation of social interaction. They like parties and may be group leaders. They have a fairly high level of energy and tend to be cheerful and optimistic. Those who know such people would describe them as active and sociable.

This person is described as being average in Neuroticism. Individuals scoring in this range are average in terms of their emotional stability. They experience a normal amount of psychological distress and have a typical balance of satisfactions and dissatisfactions with life. They are neither high nor low in self-esteem. Their ability to deal with stress is as good as the average person's.

Finally, the individual is rated in the average range in Openness. Average scorers like her value both the new and the familiar, and have an average degree of sensitivity to inner feelings. They are willing to consider new ideas on occasion, but they do not seek out novelty for its own sake.

—Detailed Interpretation: Facets of N, E, O, A, and C—

Neuroticism

This individual is perceived as being calm, relaxed, and generally free of worry. She often feels frustrated, irritable, and angry at others, but she has only the occasional periods of unhappiness that most people experience. Embarrassment or shyness when dealing with people, especially strangers, is not a problem for her. She is described as being average at controlling her impulses and desires and she is able to handle stress as well as most people.

Extraversion

This person is rated as being very warm and affectionate toward others and she usually enjoys large and noisy crowds or parties. She is seen as being forceful and dominant, preferring to be a group leader rather than a follower. The individual is described as having a high level of energy and likes to keep active and busy. Excitement, stimulation, and thrills have great appeal to her and she experiences as much joy and happiness as most women.

Openness

In experiential style, this individual is described as being somewhat open. She considers daydreaming and fantasy a waste of time, and has a limited imagination. She is like most people in her appreciation of beauty in music, art, poetry, and nature, and her feelings and emotional reactions are normal in variety and intensity. She enjoys new and different activities and has a high need for variety in her life. She has only a moderate level of intellectual curiosity and she is generally middle-of-the-road in her social, political, and moral beliefs.

Agreeableness

According to the rater, this person tends to be cynical, skeptical, and suspicious, and has a low opinion of human nature. She is described as being willing at times to flatter or trick people into doing what she wants, and she tends to put her own needs and interests before others'. This individual can be very competitive and is ready to fight for her views if necessary. She is described as quite proud of herself and her accomplishments, and happy to take credit for them. Compared to other people, she is hard-headed and tough-minded, and her social and political attitudes reflect her pragmatic realism.

Conscientiousness

This individual is perceived as being reasonably efficient and generally sensible and rational in making decisions. She is described as moderately neat, punctual, and well organized, and she is highly conscientious, adhering strictly to her ethical principles. She has a high aspiration level and strives for excellence in whatever she does. She is determined, persistent, and able to force herself to do what is necessary. She is reasonably cautious, and generally thinks things through before acting.

—Personality Correlates: Some Possible Implications—

Coping and Defenses

In coping with the stresses of everyday life, this individual is described as being not very likely to react with ineffective responses, such as hostile reactions toward others, self-blame, or escapist fantasies. She is likely to use both faith and humor in responding to threats, losses, and challenges. In addition, she is somewhat more likely to use positive thinking and direct action in dealing with problems. She is more likely to present a defensive facade of superiority than to be self-sacrificing. She may use such defense mechanisms as acting out and projection [Costa, Zonderman, & McCrae, 1991].

Somatic Complaints

This person likely responds in a normal fashion to physical problems and illness. She is prone neither to exaggerate nor to minimize physical symptoms and is fairly objective in assessing the seriousness of any medical problems that she might have.

Psychological Well-being

Although her mood and satisfaction with various aspects of her life will vary with the circumstances, in the long run this individual is likely to experience the normal course of positive and negative feelings and generally be happy. Because she is high in Conscientiousness, her accomplishments and achievements may give her greater satisfaction with life [McCrae & Costa, 1991a].

Cognitive Processes

This individual is likely to be about average in the complexity and differentiation of her thoughts, values, and moral judgments as compared to others of her level of intelligence and education. She would also probably score in the average range on measures of ego development.

Interpersonal Characteristics

Many theories propose a circular arrangement of interpersonal traits around the axes of Love and Status. Within such systems, this person would likely be described as arrogant, calculating, gregarious, sociable, and especially dominant and assured. Her traits are associated with high standing on the interpersonal dimension of Status.

Needs and Motives

Research in personality has identified a widely used list of psychological needs. Individuals differ in the degree to which these needs characterize their motivational structure. This individual is likely to show high levels of the following needs: achievement, affiliation, aggression, change, dominance, endurance (persistence), exhibition (attention), nurturance, and order. This individual is likely to show low levels of the following needs: abasement and harm avoidance (avoiding danger).

—Clinical Hypotheses: Axis II Disorders and Treatment Implications—

The NEO PI-R™ is a measure of personality traits, not psychopathology symptoms, but it is useful in clinical practice because personality profiles can suggest hypotheses about the disorders to which patients are prone and their responses to various kinds of therapy. This section of the NEO PI-R™ Interpretive Report is intended for use in clinical populations only. The hypotheses it offers should be accepted only when they are supported by other corroborating evidence.

Psychiatric diagnoses occur in men and women with different frequencies, and diagnoses are given according to uniform criteria. For that reason, information in this section of the Interpretive Report is based on Combined Gender norms. Since Same Gender Norms were used for the Interpretive Report, there may be some apparent inconsistencies in score levels and interpretations.

Axis II Disorders

Personality traits are most directly relevant to the assessment of personality disorders coded on Axis II of the *DSM-IV*. A patient may have a personality disorder in addition to an Axis I disorder, and may meet criteria for more than one personality disorder. Certain diagnoses are more common among individuals with particular personality profiles; this section calls attention to diagnoses that are likely (or unlikely) to apply.

Borderline Personality Disorder. The most common personality disorder in clinical practice is Borderline, and the mean NEO PI-R™ profile of a group of patients diagnosed as having Borderline Personality Disorder [Clarkin, Hull, Cantor, & Sanderson, 1993] provides a basis for evaluating the patient. Profile agreement between the patient and this mean profile is lower than half the subjects' in the normative sample, suggesting that the patient is unlikely to have a Borderline Personality Disorder.

Other Personality Disorders. Personality disorders can be conceptually characterized by a prototypic profile of NEO PI-R™ facets that are consistent with the definition of the disorder and its associated features [Widiger, Trull, Clarkin, Sanderson, & Costa, 1994]. The coefficient of profile agreement [McCrae, 1993] can be used to assess the overall similarity of the patient's personality to other *DSM-IV* personality disorder prototypes [McCrae, Yang, et al., 2001].

The patient's scores on N2: Angry Hostility, N4: Self-Consciousness, N6: Vulnerability, E1: Warmth, E2: Gregariousness, E4: Activity, E5: Excitement-Seeking, E6: Positive Emotions, O1: Fantasy, O3: Feelings, O4: Actions, O5: Ideas, A1: Trust, A2: Straightforwardness, A3: Altruism, C1: Competence, and C5: Self-Discipline suggest the possibility of a Histrionic Personality Disorder. [Figure 6 shows the combined profile of Case C and the Histrionic prototype.] Histrionic personality disorder is relatively common in clinical practice; the patient's coefficient of profile agreement is higher than 90% of subjects' in the normative sample.

The patient's scores on N2: Angry Hostility, N3: Depression, N4: Self-Consciousness, E1: Warmth, E3: Assertiveness, E6: Positive Emotions, O3: Feelings, O6: Values, A3: Altruism, A4: Compliance, C2: Order, C3: Dutifulness, C4: Achievement Striving, and C6: Deliberation suggest the possibility of a Obsessive-Compulsive Personality Disorder. Obsessive-Compulsive Personality Disorder is relatively common in clinical practice; the patient's coefficient of profile agreement is higher than 90% of subjects' in the normative sample.

It is unlikely that the patient has Schizoid Personality Disorder, Schizotypal Personality Disorder, Avoidant Personality Disorder, or Dependent Personality Disorder because the patient's coefficients of profile agreement are lower than 50% of the subjects' in the normative sample.

Treatment Implications

This patient scores relatively low in Neuroticism, compared to other psychotherapy patients. Her problems are likely to be due to a recent stressor or a difficult situation, and treatment may focus on dealing with those specific issues.

Because she is extraverted, this patient finds it easy to talk about her problems, and enjoys interacting with others. She is likely to respond well to forms of psychotherapy that emphasize verbal and social interactions, such as psychoanalysis and group therapy [T. Miller, 1991].

The patient scores low on Agreeableness. She is therefore likely to be skeptical and antagonistic in psychotherapy, and reluctant to establish a treatment alliance until the therapist has demonstrated his or her skill and knowledge [McCrae, Harwood, & Griffith, in press]. Individuals with extremely low levels of Agreeableness are unlikely to seek treatment voluntarily, and may terminate treatment early.

Because the patient is high in Conscientiousness, she is likely to work hard at the task of psychotherapy, arriving at therapy sessions promptly and carrying out homework assignments dutifully [Mutén, 1991]. Some evidence suggests that highly conscientious patients have better treatment outcomes.

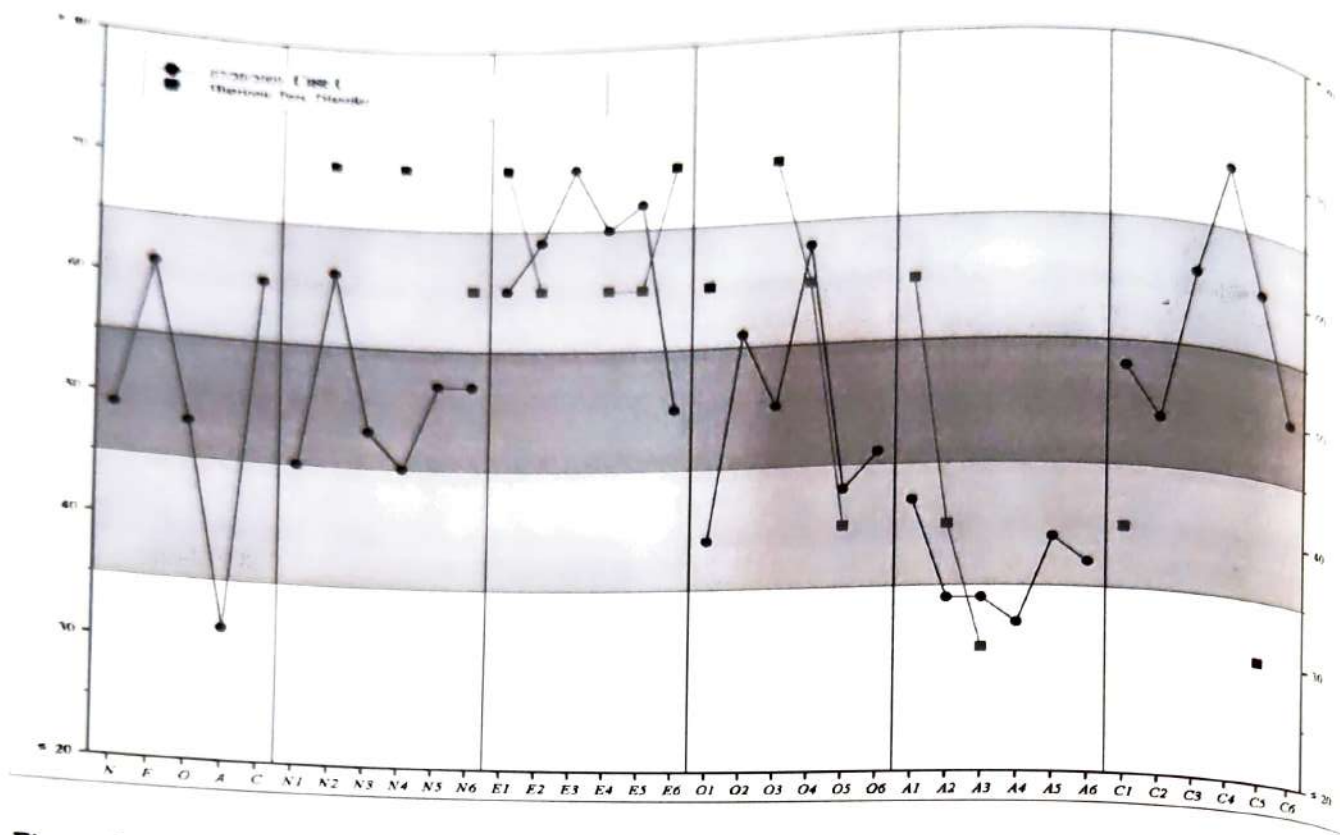


Figure 6. Combined NEO PI-R Profile Form for Case C and the Histrionic Personality Disorder prototype.

Case D

Figure 7 presents the NEO PI-R self-report profile of a 69-year-old man referred to a behavioral medicine clinic for psychophysiological treatment for chest pains and high blood pressure. This individual is a self-employed businessman who had undergone bypass surgery two years earlier and was afraid that he might die, leaving his wife and business in a vulnerable state. During the intake interview, he stated that he had a lifelong history of anxiety, especially in social situations and unfamiliar surroundings. He is well above average in intelligence.

Also shown in Figure 7 is the patient's NEO PI-R profile of Form R ratings provided by the patient's wife. Because the Form R Profile Form is used, the wife's Form R ratings are plotted directly onto the form. T scores for the patient's Form S data are obtained from the Adult Male Form S Profile Form and are plotted at the corresponding T score value on the Form R Profile Form. For example, the patient's N domain raw score on Form S was 114. According to the Adult Male Form S Profile Form, this raw score corresponds to a T score of 69. The patient's N score is plotted on the Form R Profile Form at a raw score of 108, which also corresponds to a T score of 69.

In general, the self-report and observer ratings are quite consistent at the factor level. Both ratings depict the patient as being introverted, open to experience, and conscientious. The largest discrepancies concern facets of Neuroticism, especially N4: Self-Consciousness and N6: Vulnerability. His wife clearly underestimated his standing on these facets, and discussions with the couple showed that he had been reluctant to share the extent of his distress with her. More open communication of his fears and vulnerabilities became one of the goals of his treatment. The Combined Report gives a more detailed account of agreement and disagreement, based on profile agreement statistics.

Because this patient was Very High on Openness to Experience (O), he was able to benefit from imagery and self-hypnotic techniques designed to enhance relaxation and reduce blood pressure (cf. Quails & Sheehan, 1979). Because he was High in Conscientiousness (C), he practiced these new skills regularly at home. As a result, at the end of the tenth session, he had substantially reduced his mean blood pressure and was able to discontinue therapy. The Client Report, based on his self-report and used for feedback, is given here, followed by the Combined Report.

Males Form R (Ratings)

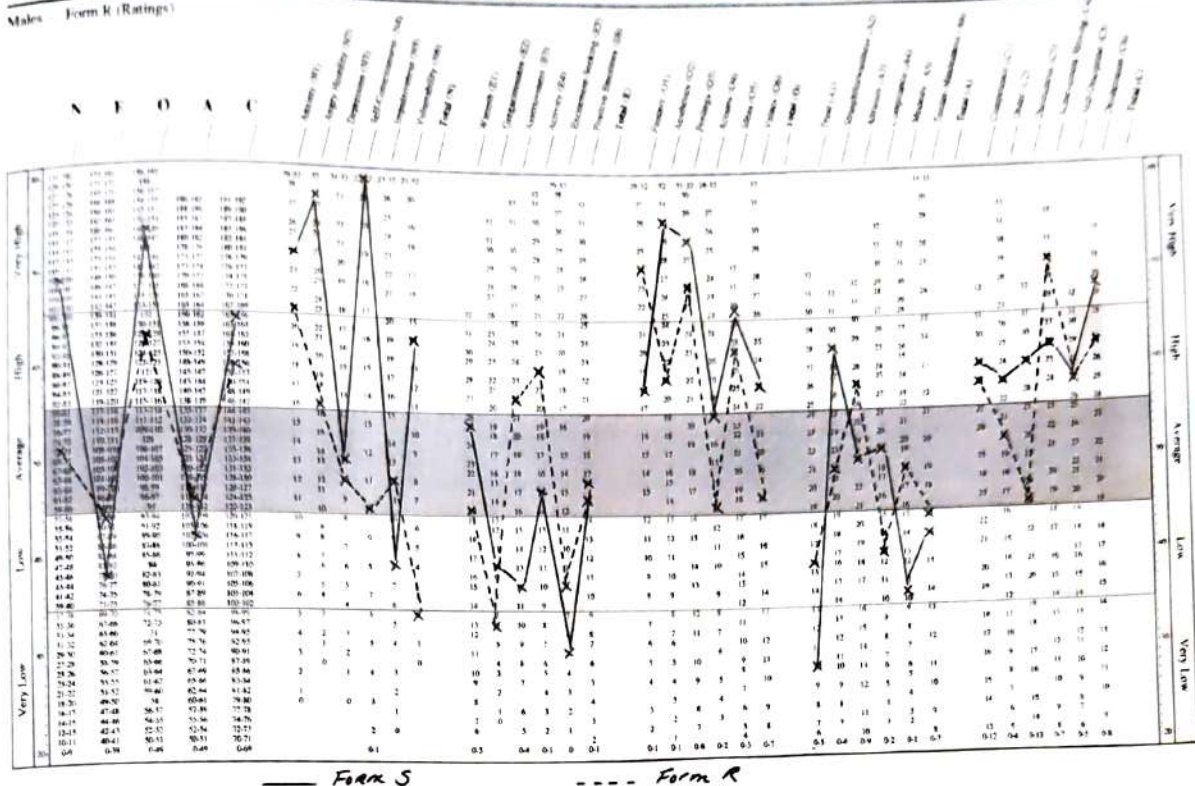


Figure 7. NEO PI-R Profile Form with self-ratings and spouse ratings for Case D. From the *Revised NEO Personality Inventory (NEO PI-R) and NEO Five-Factor Inventory (NEO-FFI) Professional Manual* (p. 27), by P. T. Costa, Jr. & R. R. McCrae, 1992, Odessa, FL: PAR. Copyright 1992 by PAR.

Your NEO PI-R Summary

The Revised NEO Personality Inventory measures five broad domains or factors of personality, and six more specific traits or factors within each domain. The responses that you gave to the statements about your thoughts, feelings, and goals can be compared to those of other adults to give a description of your personality.

The NEO Personality Inventory measures differences in personality traits. It is not a test of intelligence or ability, and is not intended to diagnose psychiatric disorders. It does, however, give you some idea about what makes you unique in your ways of thinking, feeling, and interacting with others.

This summary is intended to give you a general idea of how your personality might be described. If you completed the inventory again, you might score somewhat differently, and other people might have different views of what you are like.

If you have any questions or concerns about this summary, please feel free to discuss them with the professional who administered the inventory.

A Description of Your Personality

The N Domain

Traits in the N domain reflect different ways of reacting emotionally to distressing circumstances. Low scorers are resilient, rarely experiencing negative emotions; high scorers often have strong emotional reactions. Overall, your responses suggest that you are very high on this factor. Specifically, you are anxious, generally apprehensive, and prone to worry. You often feel frustrated, irritable, and angry at others, but you have only the occasional periods of unhappiness that most people experience. Embarrassment or shyness when dealing with people, especially strangers, is often a problem for you. You report being good at controlling your impulses and desires, but you are unable to handle stress well.

The E Domain

The E domain measures traits related to energy and enthusiasm, especially when dealing with people. Low scorers are serious and introverted; high scorers are outgoing extraverts. Your total score puts you in the very low range on this factor. You are average in your level of warmth toward others, but you rarely enjoy large and noisy crowds or parties. You are reluctant to assert yourself and prefer to stay in the background in meetings and group discussions. You have a moderate level of personal energy and an average activity level. Excitement, stimulation, and thrills have little appeal to you, but you experience as much joy and happiness as most men.

The O Domain

The facets of this domain measure responses to various kinds of experience. Low scorers are down-to-earth and conventional; they prefer the familiar and the tried-and-true. High scorers are imaginative and open-minded. You score in the very high range. Your responses suggest that you are generally open. You have a vivid imagination and an active fantasy life. You are particularly responsive to beauty as found in music, art, poetry, or nature, and your feelings and emotional reactions are varied and important to you. You sometimes enjoy new and different activities and have a moderate need for variety in your life. You are interested in intellectual challenges and in unusual ideas and perspectives and you are generally liberal in your social, political, and moral beliefs.

The A Domain

This domain is concerned with styles of interpersonal interaction. Low scorers are hard-headed and competitive; high scores compassionate and cooperative. Across the six facets in this domain, you describe yourself as being generally average on this domain. In particular, you tend to be cynical, skeptical, and suspicious, and have a low opinion of human nature. You are very candid and sincere and would find it difficult to deceive or manipulate others, and you are reasonably considerate of others and responsive to requests for help. You hold your own in conflicts with others, but you are also willing to forgive and forget. You are quite proud of yourself and your accomplishments, and happy to take credit for them. Compared to other people, you are hard-headed and tough-minded, and your social and political attitudes reflect your pragmatic realism.

The C Domain

Traits in this domain describe differences in motivation and persistence. Low scorers are easygoing and not inclined to make plans or schedules. High scorers are conscientious and well organized. Compared to other adults your score falls in the very high range on this factor. You are rational, prudent, practical, resourceful, and well-prepared. You are very neat, punctual, and well organized, and you are highly conscientious, adhering strictly to your ethical principles. You have a high aspiration level and strive for excellence in whatever you do. You are determined, persistent, and able to force yourself to do what is necessary. You are cautious and deliberate and think carefully before acting.

Combined Report

—Basis of Interpretation—

This report compares this person to other adult men. It is based on self-reports and observer ratings of the individual.

—Overall Profile Agreement—

The Coefficient of Profile Agreement, based on the factor scores of the individual and the observer, is 0.91. This means that the overall agreement on the individual's personality is very high in comparison with agreement seen among research volunteer couples.

These two profiles show substantial disagreement on the following:

Scale	Client Self-Report			Wife's Rating			Mean T Score
	Raw Score	T- Score	Range	Raw Score	T- Score	Range	
Neuroticism Facets							
(N4) Self-Consciousness	28	83	Very High	10	45	Average	66
(N6) Vulnerability	14	63	High	2	34	Very Low	48
Extraversion Facets							
(E3) Assertiveness	10	37	Low	22	57	High	47
(E4) Activity	16	47	Average	22	60	High	54
Openness Facets							
(O6) Values	24	57	High	18	46	Average	52
Agreeableness Facets							
(A4) Compliance	19	52	Average	13	40	Low	45
Conscientiousness Facets							
(C3) Dutifulness	27	60	High	23	45	Average	53

—Global Description of Personality: The Five Factors—

The most distinctive feature of this individual's personality is his standing on the factor of Conscientiousness. Men who score in this range lead very well-ordered lives, striving to meet their goals in a planful and deliberate manner. They have a high need for achievement. They are neat, punctual, and well organized, and can be relied upon to carry through on their commitments. They take moral, civic, and personal obligations quite seriously, and put business before pleasure. They have good self-discipline and have developed a number of competencies. Raters describe such people as careful, reliable, hardworking, and persevering.

This person is described as being very high in Neuroticism. Individuals scoring in this range are prone to experience a high level of negative emotion and frequent episodes of psychological distress. They are moody, overly sensitive, and dissatisfied with many aspects of their lives. They are generally low in self-esteem and may have unrealistic ideas and expectations. They are worriers who typically feel insecure about themselves and their plans. Friends and neighbors of such individuals might characterize them as nervous, self-conscious, high-strung, and vulnerable in comparison with the average person. (It is important to recall that Neuroticism is a dimension of normal personality, and high Neuroticism scores in themselves do not imply that the individual is suffering from any psychological disorder.)

Next, consider the individual's level of Openness. Very high scorers like him have a strong interest in experience for its own sake. They seek out novelty and variety, and have a marked preference for complexity. They have a heightened awareness of their own feelings and are perceptive in recognizing the emotions of others. They are very responsive to beauty in art and nature. Their attraction to new ideas and alternative value systems may make them especially tolerant of others, and may lead them to adopt unconventional attitudes. Peers rate such people as imaginative, daring, independent, and creative.

This person is described as being very low in Extraversion. Such people are quite introverted, preferring to do most things alone or with small groups of people. They avoid large, loud parties and do not enjoy meeting new people. They are usually quiet and unassertive in group interactions. They rarely experience strong positive feelings like joy or excitement. Those who know such people would probably describe them as reserved, serious, retiring, and loners. The fact that these individuals are introverted does not necessarily mean that they lack social skills—many introverts function very well in social situations, although they might prefer to avoid them. Note also that introversion does not imply introspection; these individuals are likely to be thoughtful and reflective only if they are also high in Openness.

Finally, the individual is rated in the average range in Agreeableness. People who score in this range are about as good-natured as the average person. They can be sympathetic, but can also be firm. They are trusting but not gullible, and ready to compete as well as to cooperate with others.

—Detailed Interpretation: Facets of N, E, O, A, and C—

Each of the five factors encompasses a number of more specific traits, or facets. The NEO PI-R measures six facets in each of the five factors. An examination of the facet scores provides a more detailed picture of the distinctive way that these factors are seen in this person.

Neuroticism

This individual is perceived as being anxious, generally apprehensive, and prone to worry. He often feels frustrated, irritable, and angry at others, but he has only the occasional periods of unhappiness that most people experience. Embarrassment or shyness when dealing with people, especially strangers, is often a problem for him. He is described as being good at controlling his impulses and desires and he is able to handle stress as well as most people.

Extraversion

This person is rated as being average in his level of warmth toward others, but he rarely enjoys large and noisy crowds or parties. He is as assertive as most men when the circumstances require. The individual is described as having a moderate level of personal energy and an average activity level. Excitement, stimulation, and thrills have little appeal to him, but he experiences as much joy and happiness as most men.

Openness

In experiential style, this individual is described as being generally open. He has a vivid imagination and an active fantasy life. He is particularly responsive to beauty as found in music, art, poetry, or nature, and his feelings and emotional reactions are varied and important to him. He sometimes enjoys new and different activities and has a moderate need for variety in his life. He is interested in intellectual challenges and in unusual ideas and perspectives, but he is generally middle-of-the-road in his social, political, and moral beliefs.

Agreeableness

According to the rater, this person tends to be cynical, skeptical, and suspicious, and has a low opinion of human nature. He is described as very candid and sincere and would find it difficult to deceive or manipulate others, and he is reasonably considerate of others and responsive to requests for help. This individual holds his own in conflicts with others, but he is also willing to forgive and forget. He is described as quite proud of himself and his accomplishments, and happy to take credit for them. Compared to other people, he is hard-headed and tough-minded, and his social and political attitudes reflect his pragmatic realism.

Conscientiousness

This individual is perceived as being rational, prudent, practical, resourceful, and well-prepared. He is described as very neat, punctual, and well organized, and he is reasonably dependable and reliable in meeting his obligations. He has a high aspiration level and strives for excellence in whatever he does. He is determined, persistent, and able to force himself to do what is necessary. He is cautious and deliberate and thinks carefully before acting.

—Personality Correlates: Some Possible Implications—

Coping and Defenses

In coping with the stresses of everyday life, this individual is described as being likely to react with ineffective responses, such as hostile reactions toward others, self-blame, or escapist fantasies. He is more likely than most adults to use humor and less likely to use faith in responding to threats, losses, and challenges. In addition, he is somewhat less likely to use positive thinking and direct action in dealing with problems.

Somatic Complaints

This person may be overly sensitive in monitoring and responding to physical problems and illnesses. In medical evaluations, it may be particularly important to seek objective confirmation of symptom reports where possible.

Psychological Well-being

Although his mood and satisfaction with various aspects of his life will vary with the circumstances, in the long run this individual is likely to be more sensitive to life's problems than its rewards, and so be relatively unhappy. Because he is open to experience, his moods may be more intense and varied than those of the average man. Because he is high in Conscientiousness, his accomplishments and achievements may give him greater satisfaction with life.

Cognitive Processes

This individual is likely to be more complex and differentiated in his thoughts, values, and moral judgments than others of his level of intelligence and education. He would also probably score higher on measures of ego development. Because he is open to experience, this individual is likely to perform better than average on tests of divergent thinking ability; that is, he can generate fluent, flexible, and original solutions to many problems. He may be considered creative in his work or hobbies.

Interpersonal Characteristics

Many theories propose a circular arrangement of interpersonal traits around the axes of Love and Status. Within such systems, this person would likely be described as modest, submissive, cold, unfeeling, and especially aloof and reserved. His traits are associated with low standing on the interpersonal dimensions of Love and Status.

Needs and Motives

Research in personality has identified a widely used list of psychological needs. Individuals differ in the degree to which these needs characterize their motivational structure. This individual is likely to show high levels of the following needs: achievement, aggression, cognitive structure, harm avoidance (avoiding danger), order, sentience (enjoyment of sensuous and aesthetic experiences), and understanding (intellectual stimulation). This individual is likely to show low levels of the following needs: abasement and impulsivity.

Case E

The final case study is of a 42-year-old woman who is described in the third edition of *Integrative Assessment of Adult Personality* (Harwood, Beutler, & Groth-Marnat, in press). She was evaluated by Larry E. Beutler, PhD, who then completed Form R of the NEO-PI-3. She has a long history of serious emotional and relationship problems and was clinically depressed at the time of her assessment. Her NEO-PI-3 Profile Form, with her scores compared to those of other adult men and women, is shown in Figure 8.

It should be clear at a glance that Case E is prone to many forms of psychopathology. The Global Description given in the NEO Software System Interpretive Report says, "Individuals scoring in this range [Very High Neuroticism] are prone to experience a high level of negative emotion and frequent episodes of psychological distress. They are moody, overly sensitive, and dissatisfied with many aspects of their lives."

With regard to her Very Low Conscientiousness score, the report states that "Women who score in this range have little need for achievement, putting personal interests or pleasure before business. They prefer not to make schedules, are often late for meetings and appointments, and have difficulty in finishing tasks. Their work is typically accomplished in a haphazard and disorganized fashion. They lack self-discipline, prefer play to work/school, and may seem aimless in setting goals for their lives." These statements, though likely correct, do not fully capture the seriousness of Case E's condition; the NEO Problems in

Living Checklist (NEO-PLC) can help to call attention to potential problems that may be clinically significant.

When completing the print version of the NEO-PLC, clinicians must transcribe *T* scores from the Profile Form, circle groups of relevant check boxes (e.g., all the boxes in the left-hand column under N1: Anxiety, because Case E's N1 *T* score is 72), and then determine, through a consideration of case records or subsequent interview or other assessment, which of the indicated problems is actually a significant concern for the client. Domains and their associated facets are presented in the usual NEO order; facets follow each domain, so that related topics can be covered together in follow-up interviews.

The NEO Software System expedites this process. In order to call attention to the most likely problems, it sorts domains (and facets within domain) by extremeness of *T* score (absolute deviation from $T = 50$), and it presents only those problems that are relevant to the client. For Case E, this means that the clinician will first be alerted to high Neuroticism-related problems, then to Low Conscientiousness-related problems, then to Low Extraversion and High Agreeableness-related problems. Within Neuroticism facets, problems associated with N6: Vulnerability, N3: Depression, N4: Self-Consciousness, N1: Anxiety, and N5: Impulsiveness are presented in turn. Because Case E scores in the average range on N2: Angry Hostility, problems related to that trait are omitted.

The NEO-PLC section of Case E's report begins as follows:

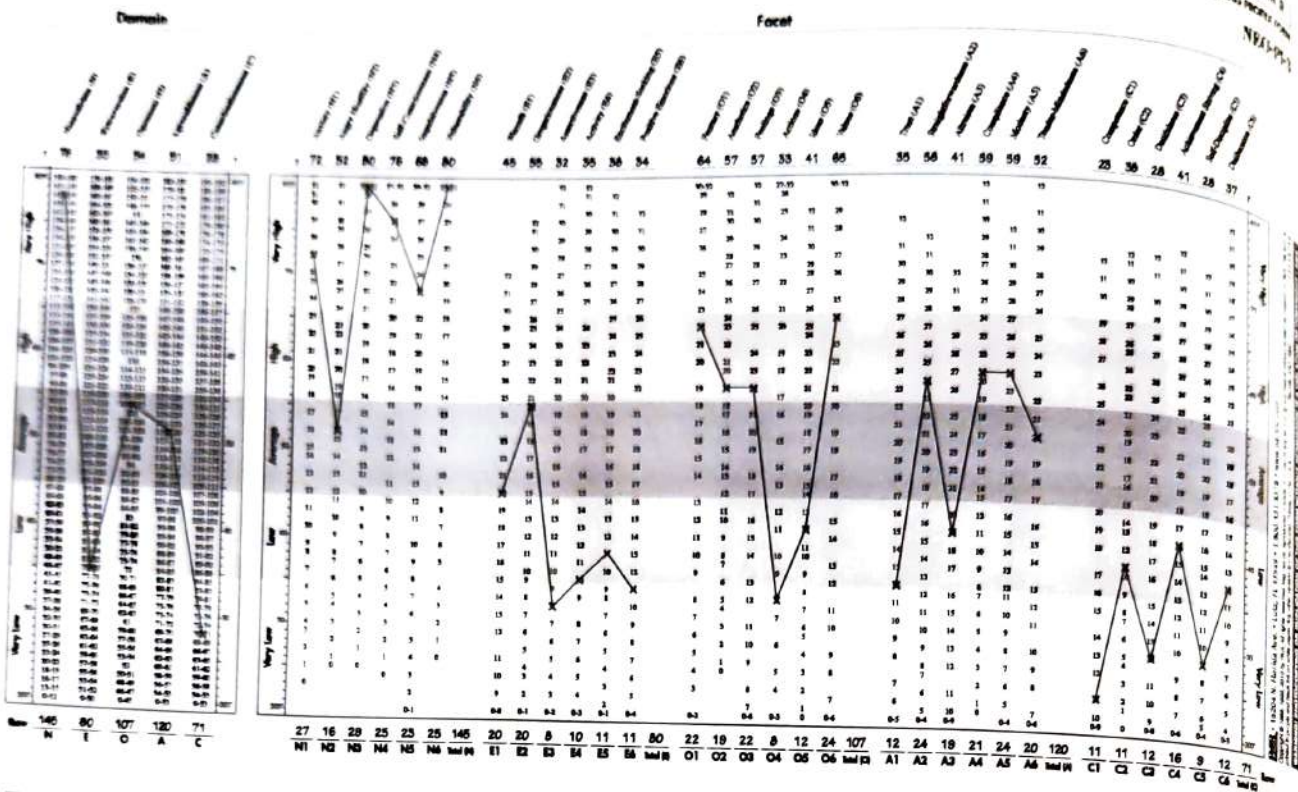


Figure 8. NEO-PI-3 Adult Combined Norms Form R Profile Form with clinician ratings for Case E. Adapted from "The NEO-PI-3," by R. R. McCrae, T. M. Harwood, & S. Griffeth, in *Integrative Assessment of Adult Personality* (3rd ed.), by T. M. Harwood, L. E. Beutler, & G. Groth-Marnat (Eds.), in press, New York, NY: Guilford Press. Copyright 2010 by Guilford Press. Adapted with permission.

—NEO Problems in Living Checklist—

Personality traits can contribute to distress and to a variety of impairments, including problems in emotional, interpersonal, experiential, attitudinal, and motivational functioning. Different traits predispose individuals to different kinds of problems. This NEO Problems in Living Checklist section provides a catalogue of potential problems based on NEO-PI-3 factor and facet scores. Factors and their facets are sorted in order of decreasing salience (as defined by absolute distance from the mean), and a list of problems relevant to this personality profile is given based on high ($T > 55$) or low ($T < 45$) scores. The clinician must then determine which, if any, of the suggested problems is clinically significant for this particular client. Pertinent problems can be checked off for future reference. *This section of the Interpretive Report is not a summary of findings about the client, but a customized guide to further inquiry.*

Neuroticism Factor

T score **78**

Women scoring in this range may experience the following problems:

- ☐ Chronic negative affects, including anxiety, fearfulness, tension, irritability, anger, dejection, hopelessness, guilt, and shame.
- ☐ Difficulty in inhibiting impulses (e.g., eating, drinking, smoking, spending money).
- ☐ Irrational beliefs (e.g., unrealistic expectations, perfectionistic demands on self, unwarranted pessimism).
- ☐ Unfounded somatic complaints.
- ☐ Helplessness and dependence on others for emotional support and decision making.

- ☐ Inability to accept criticism.
- ☐ Emotional instability; mood swings.
- ☐ Unstable relationships.

N3: Depression

T score

81

Women scoring in this range may experience the following problems:

- ☐ Suicidal thoughts.
- ☐ Chronic feelings of gloom, hopelessness, and pessimism.
- ☐ Sense of worthlessness, helplessness, and excessive guilt.
- ☐ Excessive complaints.
- ☐ Self-punitive thoughts and behaviors.
- ☐ Loneliness; perceived lack of social support.
- ☐ Lack of satisfaction or meaning in life.
- ☐ Excessive optimism and activity used to mask depression.

N6: Vulnerability

T score

80

Women scoring in this range may experience the following problems:

- ☐ Dissociative, psychotic, anxiety, or mood disorder symptomatology when experiencing stress.
- ☐ Inability to cope with stress; responds with panic, helplessness, and dismay to even minor stressors.
- ☐ Emotional instability.
- ☐ Interpersonal neediness or dependency.
- ☐ Psychosomatic complaints.

N4: Self-Consciousness

T score

76

Women scoring in this range may experience the following problems:

- ☐ Intense feelings of chagrin and embarrassment; feeling mortified, humiliated, ashamed, or disgraced in the presence of others.
- ☐ Avoidance of social situations.
- ☐ Poor social skills.
- ☐ Distorted body image; excessive concerns about body appearance.
- ☐ Sense of being an imposter.
- ☐ Speech anxiety; stage fright.

N1: Anxiety

T score

72

Women scoring in this range may experience the following problems:

- ☐ "Nerves": chronically anxious, tense, or jittery.
- ☐ Excessive worry, inhibition, and uncertainty.
- ☐ Extreme efforts to avoid danger that adversely affect decisions and actions.

N5: Impulsiveness

T score

68

Women scoring in this range may experience the following problems:

- ☐ Excessive eating, drinking, smoking, or spending.
- ☐ Susceptibility to cons, tricks, and poor business decisions.
- ☐ Poor inhibition of impulse, leading to binge eating, gambling, or excessive use of drugs and alcohol.
- ☐ Poor emotional control leading to self-mutilation or suicide attempts.
- ☐ Sexual promiscuity.
- ☐ Inability to modify behavior regardless of consequences.

Conscientiousness Factor

T score

32

Women scoring in this range may experience the following problems:

- ☐ Underachievement: not fulfilling intellectual or occupational potential.
- ☐ Poor academic performance relative to ability.
- ☐ Disregard of rules and responsibilities that can lead to trouble with the law.
- ☐ Inability to discipline self (e.g., stick to diet or exercise plan) even when required for medical reasons.
- ☐ Personal and occupational aimlessness.
- ☐ Financial problems.
- ☐ Problematic health habits that lead to medical problems.

C1: Competence

T score

23

Women scoring in this range may experience the following problems:

- ☐ Low self-esteem.
- ☐ Inability to enjoy challenges and accomplishments.
- ☐ Limited skills and underdeveloped potentials.

C3: Dutifulness

T score

28

...

The report continues with numerous other possible problems: 15 more Conscientiousness-related problems, 16 Extraversion-related problems, 17 Agreeableness-related problems, and 18 problems related to high or low facets of Openness. This long list reflects Case E's extreme standing on many traits, and serves as a warning that she is likely to be a challenging client.

Of course, even the most difficult patients will rarely have all the problems listed, and it is the clinician's task to determine which are applicable and, of those, which merit attention in treatment. Some of these determinations can be done on the basis of the clinician's knowledge of the patient's history. For example, Case E recently attempted suicide (see McCrae, Harwood, & Griffeth, in press), so, under N3: Depression problems

- ☒ Suicidal thoughts.
- ☒ Self-punitive thoughts and behaviors.

should be checked off. Because Case E has a history of alcohol abuse, the clinician should check off the N5: Impulsiveness problem

- ☒ Poor inhibition of impulse, leading to binge eating, gambling, excessive use of drugs and alcohol.

In addition, Case E's life history reveals that, as an adolescent, she had conflicts with her parents because of their rigid religious beliefs, so the O6: Values problem

- ☒ Rebellious rejection of conventionality; defiance of cultural norms.

could also be checked off. A clinician familiar with Case E would also be able to rule out a few items immediately, and they could be crossed off the list—for example:

- ☐ ~~Excessive optimism and activity used to mask depression.~~

After this initial review, the clinician should examine the remaining items and determine (a) which of these require more information from the patient (or other informants) and (b) which of these in fact merit additional assessment. With seriously troubled patients like Case E, many of the lesser problems—such as the O4: Actions problem,

- ☐ Inability to adapt to change and technological innovation.

may need to be ignored; however, in a college counseling setting, such problems might be an appropriate focus of concern. The clinician can indicate with a question mark on the form which problems should be pursued—for example:

- ☒ Limited skills and underdeveloped potentials.

The clinician then can use these marked items as a guide to a subsequent clinical interview. When a final list of applicable and clinically relevant problems has been identified, it can serve as a basis for treatment planning and the periodic evaluation of progress.

A major advantage of using the NEO-PLC is that it may call attention to underlying or contextual problems that the

patient does not volunteer as part of the presenting problem, and that the clinician may not otherwise happen to consider. For example, Case E is high on Agreeableness, which is usually desirable, because it facilitates the therapeutic alliance (see T. Miller, 1991). But the NEO-PLC lists a number of problems associated with high scores on Agreeableness facets. Case E's high Compliance score suggests that she might experience

- ☒ Exploitation or victimization due to a failure to protect or defend self.

In fact, that has been a recurrent feature of her close interpersonal relationships (McCrae, Harwood, et al., in press). Reading through the entire list of potential problems associated with a NEO-PI-3 profile can give the clinician a sense of the patient's vulnerabilities and of problems that may arise in the future.